

# Linear Spin-Wave Theory: Draft Documents for Physics Review

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# Paraunitary Diagonalization

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The quadratic Hamiltonian  $\hat{H}_2$  of *Momentum-Space BdG Hamiltonian* couples the boson  $\hat{a}_{\mathbf{k}\mu}$  to  $\hat{a}_{-\mathbf{k}\nu}^\dagger$  through the anomalous block  $B_{\mathbf{k}}$ . A linear transformation that mixes these operators separates  $\hat{H}_2$  into independent magnon modes. Because the new operators must again be bosons, the transformation preserves the Nambu metric  $\Sigma_3$  rather than the identity, and is paraunitary rather than unitary. The sections below derive the paraunitary condition, the particle–hole structure of the magnon spectrum, the diagonal magnon Hamiltonian with its zero-point correction, and Colpa’s construction of the transformation. The construction requires  $H_{\mathbf{k}}$  to be positive definite; the last section describes what changes when it is not.

## Bogoliubov Transformation and the Paraunitary Condition

We write the Nambu spinor of Holstein–Primakoff bosons,  $\hat{\Psi}_{\mathbf{k}}$ , as a linear combination of the magnon spinor  $\hat{\Phi}_{\mathbf{k}}$  defined in *Notation and Conventions*,

$$\hat{\Psi}_{\mathbf{k}} = T_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} : \quad \begin{pmatrix} \hat{a}_{\mathbf{k}} \\ \hat{a}_{-\mathbf{k}}^\dagger \end{pmatrix} = \begin{pmatrix} P_{\mathbf{k}} & Q_{-\mathbf{k}} \\ Q_{\mathbf{k}}^* & P_{-\mathbf{k}}^* \end{pmatrix} \begin{pmatrix} \hat{b}_{\mathbf{k}} \\ \hat{b}_{-\mathbf{k}}^\dagger \end{pmatrix}, \quad (1)$$

where  $P_{\mathbf{k}}$  and  $Q_{\mathbf{k}}$  are  $N_{\text{sub}} \times N_{\text{sub}}$  matrices. The first row reads  $\hat{a}_{\mathbf{k}\mu} = \sum_n [(P_{\mathbf{k}})_{\mu n} \hat{b}_{\mathbf{k}n} + (Q_{-\mathbf{k}})_{\mu n} \hat{b}_{-\mathbf{k}n}^\dagger]$ . The second row is the Hermitian conjugate of the first row at  $-\mathbf{k}$ , so the block structure of  $T_{\mathbf{k}}$  follows from the requirement that the lower components of  $\hat{\Psi}_{\mathbf{k}}$  are the adjoints of the upper components of  $\hat{\Psi}_{-\mathbf{k}}$ .

Both spinors obey the same bosonic commutation relations. For the components of  $\hat{\Psi}_{\mathbf{k}}$  these relations read  $[(\hat{\Psi}_{\mathbf{k}})_{p'}, (\hat{\Psi}_{\mathbf{k}}^\dagger)_q] = (\Sigma_3)_{pq}$ , since the lower components are creation operators. Requiring the same relations for  $\hat{\Phi}_{\mathbf{k}}$  and inserting the transformation gives  $[(\hat{\Psi}_{\mathbf{k}})_{p'}, (\hat{\Psi}_{\mathbf{k}}^\dagger)_q] = (T_{\mathbf{k}} \Sigma_3 T_{\mathbf{k}}^\dagger)_{pq}$ , so the transformation must satisfy

$$T_{\mathbf{k}} \Sigma_3 T_{\mathbf{k}}^\dagger = \Sigma_3 \quad \Longleftrightarrow \quad T_{\mathbf{k}}^\dagger \Sigma_3 T_{\mathbf{k}} = \Sigma_3. \quad (2)$$

In blocks, the first form contains the four conditions  $P_{\mathbf{k}} P_{\mathbf{k}}^\dagger - Q_{-\mathbf{k}} Q_{-\mathbf{k}}^\dagger = I$ ,  $P_{\mathbf{k}} Q_{\mathbf{k}}^\dagger - Q_{-\mathbf{k}} P_{-\mathbf{k}}^\dagger = 0$ , and their conjugate counterparts. The two forms are equivalent because either one gives the inverse  $T_{\mathbf{k}}^{-1} = \Sigma_3 T_{\mathbf{k}}^\dagger \Sigma_3$ . A unitary matrix preserves  $I$  instead of  $\Sigma_3$ ; a paraunitary  $T_{\mathbf{k}}$  is unitary only when  $Q_{\mathbf{k}} = 0$ , that is, when  $\hat{H}_2$  conserves the boson number.

## Diagonal Form and the Particle–Hole Structure of the Spectrum

We seek a paraunitary  $T_k$  for which  $D_k = T_k^\dagger H_k T_k$  is diagonal. With the inverse above, this condition is equivalent to an eigenvalue problem:

$$T_k^{-1} \Sigma_3 H_k T_k = \Sigma_3 T_k^\dagger H_k T_k = \Sigma_3 D_k.$$

The columns of  $T_k$  are therefore eigenvectors of the non-Hermitian matrix  $\Sigma_3 H_k$ , and the signed diagonal entries  $\lambda_{pk} = (\Sigma_3)_{pp} (D_k)_{pp}$  are its eigenvalues.

The BdG matrix relates  $k$  and  $-k$ . With  $\Sigma_1 = \sigma_1 \otimes I_{N_{\text{sub}}}$ , which exchanges the particle and hole blocks, the block relations  $A_k^\dagger = A_k$  and  $B_{-k} = B_k^\dagger$  give

$$\Sigma_1 H_{-k}^* \Sigma_1 = H_k. \quad (3)$$

Since  $\Sigma_3 \Sigma_1 = -\Sigma_1 \Sigma_3$ , an eigenvector  $v$  of  $\Sigma_3 H_{-k}$  with eigenvalue  $\lambda$  yields the eigenvector  $\Sigma_1 v^*$  of  $\Sigma_3 H_k$  with eigenvalue  $-\lambda^*$ . For a positive-definite  $H_k$ , the section on Colpa's construction shows that the eigenvalues are real,  $N_{\text{sub}}$  of them positive and  $N_{\text{sub}}$  negative. We denote the positive eigenvalues at  $k$  by the magnon energies  $\varepsilon_{nk}$ ; the negative eigenvalues at  $k$  are then  $-\varepsilon_{n,-k}$ . Ordering the positive eigenvalues first, and the negative eigenvalues in the band order at  $-k$ , the diagonal form is

$$T_k^\dagger H_k T_k = \text{diag}(\varepsilon_{1k}, \dots, \varepsilon_{N_{\text{sub}}k}, \varepsilon_{1,-k}, \dots, \varepsilon_{N_{\text{sub}},-k}). \quad (4)$$

The hole columns of  $T_k$  can be chosen as  $\Sigma_1$  times the complex conjugates of the particle columns of  $T_{-k}$ , which reproduces the block structure of Eq. (1). The energies  $\varepsilon_{nk}$  and  $\varepsilon_{n,-k}$  need not coincide; they coincide when a symmetry of the model and the ordered state maps  $k$  to  $-k$ .

## Diagonal Magnon Hamiltonian and Zero-Point Correction

Inserting  $\hat{\Psi}_k = T_k \hat{\Phi}_k$  into  $\hat{H}_2 = \frac{1}{2} \sum_k (\hat{\Psi}_k^\dagger H_k \hat{\Psi}_k - \text{Tr } A_k)$  and using Eq. (4) gives

$$\hat{H}_2 = \frac{1}{2} \sum_{k \in \text{MBZ}} \sum_{n=1}^{N_{\text{sub}}} (\varepsilon_{nk} \hat{b}_{kn}^\dagger \hat{b}_{kn} + \varepsilon_{n,-k} \hat{b}_{-kn} \hat{b}_{-kn}^\dagger) - \frac{1}{2} \sum_{k \in \text{MBZ}} \text{Tr } A_k.$$

The commutator  $\hat{b}_{-kn} \hat{b}_{-kn}^\dagger = \hat{b}_{-kn}^\dagger \hat{b}_{-kn} + 1$  and the relabeling  $-k \rightarrow k$ , allowed because the  $N_{\text{uc}}$  momenta of the MBZ form a set closed under inversion modulo reciprocal-lattice vectors, turn this expression into

$$\hat{H}_2 = \sum_{k \in \text{MBZ}} \sum_{n=1}^{N_{\text{sub}}} \varepsilon_{nk} \hat{b}_{kn}^\dagger \hat{b}_{kn} + \Delta E_{\text{zp}}, \quad \Delta E_{\text{zp}} = \frac{1}{2} \sum_{k \in \text{MBZ}} \left( \sum_{n=1}^{N_{\text{sub}}} \varepsilon_{nk} - \text{Tr } A_k \right). \quad (5)$$

Each magnon mode  $(k, n)$  is an independent harmonic oscillator of energy  $\varepsilon_{nk}$ . The magnon vacuum, annihilated by every  $\hat{b}_{kn}$ , is the ground state of  $\hat{H}_2$ , and the LSWT ground-state energy is

$$E_{\text{GS}} = E_{\text{cl}} + \Delta E_{\text{zp}}, \quad \mathcal{E}_{\text{GS}} = \frac{E_{\text{GS}}}{N_{\text{site}}}, \quad N_{\text{site}} = N_{\text{uc}} N_{\text{sub}}. \quad (6)$$

The zero-point correction  $\Delta E_{\text{zp}}$  is the energy of the magnon vacuum relative to the classical energy  $E_{\text{cl}}$ ; it combines the zero-point energies  $\frac{1}{2}\varepsilon_{nk}$  of the oscillators with the trace subtraction in  $\hat{H}_2$ . The trace subtraction removes the constant  $\frac{1}{2}\sum_{\mathbf{k}} \text{Tr} A_{-\mathbf{k}}$  that appears when the normal-ordered quadratic Hamiltonian is rewritten in the symmetric Nambu form, which commutes the hole components  $\hat{a}_{-\mathbf{k}}\hat{a}_{-\mathbf{k}}^\dagger$ ; replacing  $A_{-\mathbf{k}}$  by  $A_{\mathbf{k}}$  in this constant uses the same inversion closure as above. The normalization of the subtraction relative to the real-space construction, including any constant from Holstein–Primakoff terms that are not normal ordered, is still provisional in *Momentum-Space BdG Hamiltonian*;  $\Delta E_{\text{zp}}$  inherits that condition. Because  $\text{Tr} A_{\mathbf{k}}$  is real and the momentum set is closed under inversion,  $\sum_{\mathbf{k}} \text{Tr} A_{\mathbf{k}} = \frac{1}{2}\sum_{\mathbf{k}} \text{Tr} H_{\mathbf{k}}$ , so the correction can equally be written as  $\frac{1}{2}\sum_{\mathbf{k}} \sum_n \varepsilon_{nk} - \frac{1}{4}\sum_{\mathbf{k}} \text{Tr} H_{\mathbf{k}}$ . The two forms agree after the momentum sum; their summands need not agree at a single momentum.

## Colpa’s Construction for a Positive-Definite BdG Matrix

A paraunitary  $T_{\mathbf{k}}$  that brings  $H_{\mathbf{k}}$  to a diagonal form with positive entries exists if and only if  $H_{\mathbf{k}}$  is positive definite. If such a  $T_{\mathbf{k}}$  exists,  $H_{\mathbf{k}} = (T_{\mathbf{k}}^{-1})^\dagger D_{\mathbf{k}} T_{\mathbf{k}}^{-1}$  is congruent to a positive diagonal matrix and is therefore positive definite. Conversely, Colpa’s construction produces  $T_{\mathbf{k}}$  from a positive-definite  $H_{\mathbf{k}}$ . We suppress the momentum index in the rest of this section.

A positive-definite  $H$  has a Cholesky factorization  $H = KK^\dagger$  with an invertible lower-triangular  $K$ . The Hermitian matrix  $K^\dagger \Sigma_3 K$  is congruent to  $\Sigma_3$ , so by Sylvester’s law of inertia it has  $N_{\text{sub}}$  positive and  $N_{\text{sub}}$  negative eigenvalues. We diagonalize it with a unitary matrix  $V$ ,

$$V\Lambda V^\dagger = K^\dagger \Sigma_3 K, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_{2N_{\text{sub}}}),$$

ordering the eigenvalues so that the positive ones come first. The diagonal matrix  $\Lambda \Sigma_3$  then has only positive entries, and the transformation is

$$T = (K^\dagger)^{-1} V (\Lambda \Sigma_3)^{1/2}. \quad (7)$$

Both requirements follow from the definitions of  $K$  and  $V$ , using that the diagonal matrices  $\Lambda$ ,  $\Sigma_3$ , and  $(\Lambda \Sigma_3)^{1/2}$  commute and that  $(\Lambda \Sigma_3)^{1/2}$  is real:

$$\begin{aligned} T \Sigma_3 T^\dagger &= (K^\dagger)^{-1} V (\Lambda \Sigma_3)^{1/2} \Sigma_3 (\Lambda \Sigma_3)^{1/2} V^\dagger K^{-1} = (K^\dagger)^{-1} V \Lambda V^\dagger K^{-1} = \Sigma_3, \\ T^\dagger H T &= (\Lambda \Sigma_3)^{1/2} V^\dagger K^{-1} K K^\dagger (K^\dagger)^{-1} V (\Lambda \Sigma_3)^{1/2} = \Lambda \Sigma_3. \end{aligned}$$

The eigenvalues  $\lambda_p$  are those of  $\Sigma_3 H$ , because  $\Sigma_3 H = \Sigma_3 K K^\dagger$  is similar to  $K^\dagger \Sigma_3 K$  through  $K^\dagger$ . They are the signed energies of the previous section,  $\varepsilon_{nk}$  on the particle columns and  $-\varepsilon_{n,-\mathbf{k}}$  on the hole columns, and  $\Lambda \Sigma_3$  is the diagonal matrix of Eq. (4). This also shows that the eigenvalues of  $\Sigma_3 H_{\mathbf{k}}$  are real when  $H_{\mathbf{k}}$  is positive definite.

The construction fixes  $T_{\mathbf{k}}$  only up to  $T_{\mathbf{k}} \rightarrow T_{\mathbf{k}} U$ , where  $U$  is paraunitary and commutes with  $\Sigma_3 D_{\mathbf{k}}$ . Such a  $U$  is block diagonal in the eigenspaces of the signed energies, which do not mix particle and hole columns when all energies are positive, and is unitary within each block. It therefore multiplies each column by a phase and mixes columns only within degenerate bands. The energies and  $\Delta E_{\text{zp}}$  do not depend on this freedom; the Berry connection of *Topological Magnon Quantities* does. Hole

columns obtained independently at  $k$  agree with the particle–hole partners of the particle columns at  $-k$  up to the same freedom.

## Positive-Semidefinite and Indefinite BdG Matrices

The quadratic form  $\hat{\Psi}_k^\dagger H_k \hat{\Psi}_k$  is the harmonic energy of the spin fluctuations about the classical reference configuration of *Classical Order and Local Frame*. Positive definiteness of  $H_k$  at every momentum means that every harmonic fluctuation raises the energy. If  $H_k$  is indefinite at some momentum, a fluctuation lowers the energy, the reference configuration is not a local minimum, and the magnon vacuum is not the ground state of  $\hat{H}_2$ . The Cholesky factorization then fails. The matrix  $\Sigma_3 H_k$  may have complex eigenvalues, which signal a dynamical instability, or real eigenvalues with a mode of negative energy, as for a ferromagnet polarized against the field; in either case the harmonic expansion does not describe the equilibrium state.

A positive-semidefinite  $H_k$  has zero modes, for example Goldstone modes of a broken continuous symmetry. Its Cholesky factor is singular, and Colpa's construction does not apply. A single mode with real blocks  $A = A$  and  $B = B$ , for which  $\varepsilon = \sqrt{A^2 - B^2}$ , shows the two possible behaviors at  $\varepsilon = 0$ :

- For  $A = B > 0$ ,  $H$  is positive semidefinite and  $\Sigma_3 H = A \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$  satisfies  $(\Sigma_3 H)^2 = 0$ . The matrix is not diagonalizable, and no paraunitary  $T$  exists. The Hamiltonian  $\frac{A}{2}(\hat{a} + \hat{a}^\dagger)^2$  contains one canonical coordinate but not its conjugate momentum, so the mode is not a harmonic oscillator. A Goldstone mode of an antiferromagnet at  $k = 0$  has this form.
- For  $A = B = 0$ ,  $\Sigma_3 H = 0$  is diagonalizable,  $T = I$  is paraunitary, and the mode is a boson of zero energy. The Goldstone mode of a Heisenberg ferromagnet at  $k = 0$  and zero field has this form.

Adding a small positive shift  $\delta I_{2N_{\text{sub}}}$  makes  $H_k$  positive definite. The energies then converge as  $\delta \rightarrow 0$ , but in the first case the entries of  $T$  grow as  $\delta^{-1/4}$ , since  $\varepsilon = \sqrt{2A\delta + \delta^2}$  while the diagonal entries of  $T$  scale as  $(A/\varepsilon)^{1/2}$ . Zero modes usually occur at isolated momenta, where  $\varepsilon_{nk} \rightarrow 0$  continuously; they do not change  $\Delta E_{\text{zp}}$ . Quantities weighted by  $1/\varepsilon_{nk}$  or by the Bose–Einstein distribution, such as magnon numbers and correlation functions, require separate treatment and are discussed in the observable documents. A complete diagonalization theory for positive-semidefinite quadratic boson Hamiltonians, which brings them to a standard form that includes such zero modes, is given by Colpa (1986).

## References

### Internal Documents

- *Notation and Conventions*: defines the Nambu spinors  $\hat{\Psi}_k$  and  $\hat{\Phi}_k$ , the metric  $\Sigma_3$ , the paraunitary matrix  $T_k$ , and the energy symbols  $\varepsilon_{nk}$ ,  $E_{\text{cl}}$ ,  $\Delta E_{\text{zp}}$ , and  $E_{\text{GS}}$ .
- *Momentum-Space BdG Hamiltonian*: supplies  $\hat{H}_2$ , the BdG blocks  $A_k$  and  $B_k$ , their block relations, and the provisional trace subtraction.
- *Classical Order and Local Frame*: defines the classical reference configuration about which  $H_k$  is expanded.
- *Paraunitarity Proofs*: is reserved for extended metric-preservation proofs.
- *Magnon Observables*: uses the magnon energies and the transformation  $T_k$ .
- *Topological Magnon Quantities*: uses the columns of  $T_k$  and their gauge freedom in the Berry curvature.

## External Sources

- J. H. P. Colpa, “Diagonalization of the Quadratic Boson Hamiltonian,” *Physica A* **93**, 327–353 (1978), [doi:10.1016/0378-4371\(78\)90160-7](https://doi.org/10.1016/0378-4371(78)90160-7): paraunitary diagonalization of a positive-definite quadratic boson Hamiltonian through the Cholesky factorization; the construction of Eq. (7).
- J. H. P. Colpa, “Diagonalization of the Quadratic Boson Hamiltonian with Zero Modes,” *Physica A* **134**, 377–416 (1986), [doi:10.1016/0378-4371\(86\)90056-7](https://doi.org/10.1016/0378-4371(86)90056-7), and **134**, 417–442 (1986), [doi:10.1016/0378-4371\(86\)90057-9](https://doi.org/10.1016/0378-4371(86)90057-9): standard form of positive-semidefinite quadratic boson Hamiltonians with zero modes.

# Appendix: Paraunitarity Proofs

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The diagonalization of a bosonic Bogoliubov–de Gennes (BdG) matrix rests on a few algebraic facts about the indefinite metric  $\Sigma_3 = \sigma_3 \otimes I_{N_{\text{sub}}}$ : paraunitary matrices form a group, the eigenvectors of  $\Sigma_3 H$  are orthogonal with respect to  $\Sigma_3$ , and positive definiteness of  $H$  fixes both the reality of the spectrum and the sign of each eigenvector's  $\Sigma_3$ -norm. This appendix proves these facts and the properties of the transformation  $T_k$  used in *Paraunitary Diagonalization*. Throughout,  $H$  is a Hermitian  $2N_{\text{sub}} \times 2N_{\text{sub}}$  matrix, and the momentum index is suppressed where only one momentum enters.

## Paraunitary Matrices Form a Group

A matrix  $T$  is paraunitary when  $T \Sigma_3 T^\dagger = \Sigma_3$ . Taking determinants gives  $|\det T|^2 = 1$ , so  $T$  is invertible. Multiplying the condition by  $\Sigma_3$  on the right gives  $T(\Sigma_3 T^\dagger \Sigma_3) = I$ , so

$$T^{-1} = \Sigma_3 T^\dagger \Sigma_3.$$

Inserting this inverse into  $T^{-1}T = I$  gives  $\Sigma_3 T^\dagger \Sigma_3 T = I$ , which is the second form  $T^\dagger \Sigma_3 T = \Sigma_3$ . The argument runs equally in the other direction, so the two forms of the paraunitary condition are equivalent. If  $T_1$  and  $T_2$  are paraunitary, then  $T_1 T_2 \Sigma_3 T_2^\dagger T_1^\dagger = T_1 \Sigma_3 T_1^\dagger = \Sigma_3$ , and the inverse  $\Sigma_3 T_1^\dagger \Sigma_3$  satisfies the second form of the condition. Paraunitary matrices therefore form a group, the pseudo-unitary group  $U(N_{\text{sub}}, N_{\text{sub}})$ . A paraunitary matrix that is also unitary commutes with  $\Sigma_3$  and is block diagonal in the particle and hole components.

## Metric Orthogonality of Eigenvectors

Let  $\Sigma_3 H v = \lambda v$  and  $\Sigma_3 H w = \mu w$ . Multiplying the first relation by  $w^\dagger \Sigma_3$  and the conjugate of the second by  $v$ , and using  $\Sigma_3^2 = I$  and  $H^\dagger = H$ , gives

$$w^\dagger H v = \lambda w^\dagger \Sigma_3 v = \mu^* w^\dagger \Sigma_3 v, \quad (\lambda - \mu^*) w^\dagger \Sigma_3 v = 0. \quad (8)$$

Eigenvectors whose eigenvalues satisfy  $\lambda \neq \mu^*$  are therefore orthogonal in the indefinite inner product  $w^\dagger \Sigma_3 v$ . For  $w = v$  the relation reads  $v^\dagger H v = \lambda v^\dagger \Sigma_3 v$ , and an eigenvector with a nonreal eigenvalue has  $v^\dagger \Sigma_3 v = 0$ .

## Consequences of Positive Definiteness

Assume that  $H$  is positive definite. Three properties of  $\Sigma_3 H$  follow.

**Real, nonzero eigenvalues with a fixed metric sign.** For an eigenvector  $v$ , the relation  $v^\dagger H v = \lambda v^\dagger \Sigma_3 v$  has a positive left-hand side. If  $\lambda$  were not real, the previous section would give  $v^\dagger \Sigma_3 v = 0$  and hence  $v^\dagger H v = 0$ , a contradiction. Therefore  $\lambda$  is real and nonzero,  $v^\dagger \Sigma_3 v \neq 0$ , and

$$\text{sign } \lambda = \text{sign } (v^\dagger \Sigma_3 v). \quad (9)$$

Eigenvectors with positive eigenvalues have a positive  $\Sigma_3$ -norm and become particle columns of  $T$ ; eigenvectors with negative eigenvalues have a negative  $\Sigma_3$ -norm and become hole columns. For an indefinite  $H$  this link fails: an eigenvector with a positive eigenvalue can have a negative  $\Sigma_3$ -norm, which is the negative-energy mode of the indefinite case in *Paraunitary Diagonalization*.

**Diagonalizability.** The Cholesky factorization  $H = K K^\dagger$  makes  $\Sigma_3 H = \Sigma_3 K K^\dagger$  similar, through  $K^\dagger$ , to the Hermitian matrix  $K^\dagger \Sigma_3 K$ :

$$K^\dagger (\Sigma_3 H) (K^\dagger)^{-1} = K^\dagger \Sigma_3 K.$$

A Hermitian matrix is diagonalizable with real eigenvalues, so  $\Sigma_3 H$  is diagonalizable with the same eigenvalues. Jordan blocks, such as the one of the antiferromagnetic Goldstone mode, can occur only when  $H$  is singular.

**Equal numbers of positive and negative eigenvalues.** The matrix  $K^\dagger \Sigma_3 K$  is congruent to  $\Sigma_3$  through the invertible matrix  $K$ . By Sylvester's law of inertia, congruent Hermitian matrices have the same numbers of positive and negative eigenvalues, so  $\Sigma_3 H$  has  $N_{\text{sub}}$  positive and  $N_{\text{sub}}$  negative eigenvalues.

## Existence of the Paraunitary Transformation

The three properties construct  $T$  directly from eigenvectors. Within an eigenspace of  $\Sigma_3 H$  with eigenvalue  $\lambda$ , the form  $w^\dagger \Sigma_3 v$  is definite with the sign of  $\lambda$ , because Eq. (9) holds for every vector of the eigenspace. Gram–Schmidt orthogonalization with respect to this definite form gives a basis with  $v^\dagger \Sigma_3 v = \text{sign } \lambda$ . Eigenvectors of different eigenvalues are already  $\Sigma_3$ -orthogonal by Eq. (8), since all eigenvalues are real. Collecting the  $N_{\text{sub}}$  vectors of positive eigenvalues first and the  $N_{\text{sub}}$  vectors of negative eigenvalues second gives a matrix  $T$  with  $T^\dagger \Sigma_3 T = \Sigma_3$ . The eigenvalue equation  $\Sigma_3 H T = T \Sigma_3 D$ , with  $D$  the diagonal matrix of the absolute values of the eigenvalues, then gives  $T^\dagger H T = T^\dagger \Sigma_3 T \Sigma_3 D = D$ , a diagonal matrix with positive entries.

Conversely, if a paraunitary  $T$  with  $T^\dagger H T = D$  positive and diagonal exists, then  $H = (T^{-1})^\dagger D T^{-1}$  is congruent to a positive diagonal matrix and is positive definite. A paraunitary transformation to a diagonal form with positive entries therefore exists if and only if  $H$  is positive definite.

## Colpa's Transformation Satisfies Both Conditions

Colpa's construction,  $T = (K^\dagger)^{-1} V (\Lambda \Sigma_3)^{1/2}$  with  $V \Lambda V^\dagger = K^\dagger \Sigma_3 K$ , is verified in *Paraunitary Diagonalization* through  $T \Sigma_3 T^\dagger = \Sigma_3$  and  $T^\dagger H T = \Lambda \Sigma_3$ . The second form of the paraunitary condition can also be checked directly. The matrix  $K^\dagger \Sigma_3 K$  is invertible, so  $\Lambda$  is invertible, and  $\Sigma_3^{-1} = \Sigma_3$  gives  $K^{-1} \Sigma_3 (K^\dagger)^{-1} = (K^\dagger \Sigma_3 K)^{-1} = V \Lambda^{-1} V^\dagger$ . Hence

$$T^\dagger \Sigma_3 T = (\Lambda \Sigma_3)^{1/2} V^\dagger K^{-1} \Sigma_3 (K^\dagger)^{-1} V (\Lambda \Sigma_3)^{1/2} = (\Lambda \Sigma_3)^{1/2} \Lambda^{-1} (\Lambda \Sigma_3)^{1/2} = \Sigma_3,$$

where the last step uses that the diagonal matrices commute and  $\Lambda \Sigma_3 \Lambda^{-1} = \Sigma_3$ . This is the check left in commented form in the restructured TeX appendix; by the first section it is equivalent to  $T \Sigma_3 T^\dagger = \Sigma_3$ .

## Freedom in the Transformation

Let  $T$  and  $T'$  both be paraunitary with  $T^\dagger H T = T'^\dagger H T' = D$ , with the same ordering of the diagonal entries. The matrix  $U = T^{-1} T'$  is paraunitary by the group property. From  $\Sigma_3 H T = T \Sigma_3 D$  and the same relation for  $T'$ ,

$$U \Sigma_3 D = T^{-1} \Sigma_3 H T' = \Sigma_3 D U,$$

so  $U$  commutes with the diagonal matrix  $\Sigma_3 D$  of signed energies and is block diagonal in its eigenspaces. When  $H$  is positive definite, each eigenspace contains only particle columns or only hole columns, because their signed energies have opposite signs. On such a block  $\Sigma_3$  is  $+I$  or  $-I$ , and the paraunitary condition restricted to the block is unitarity. The transformation is therefore unique up to a phase for each nondegenerate column and a unitary mixing within each degenerate set of columns.

## Particle–Hole Structure of the Transformation

The Bogoliubov transformation has the block form  $T_k = \begin{pmatrix} P_k & Q_{-k} \\ Q_k^* & P_{-k}^* \end{pmatrix}$ , so that the lower components of  $\hat{\Psi}_k$  are the adjoints of the upper components of  $\hat{\Psi}_{-k}$ . With  $\Sigma_1 = \sigma_1 \otimes I_{N_{\text{sub}}}$ , which exchanges the two block rows when applied on the left and the two block columns when applied on the right, the block form is equivalent to

$$T_k = \Sigma_1 T_{-k}^* \Sigma_1. \quad (10)$$

A transformation of this form can always be chosen. Let  $t$  be a particle column of  $T_{-k}$  with  $\Sigma_3 H_{-k} t = \varepsilon t$  and  $t^\dagger \Sigma_3 t = 1$ . The particle–hole relation  $\Sigma_1 H_{-k}^* \Sigma_1 = H_k$  and  $\Sigma_3 \Sigma_1 = -\Sigma_1 \Sigma_3$  give  $\Sigma_3 H_k (\Sigma_1 t^*) = -\varepsilon \Sigma_1 t^*$ , and

$$(\Sigma_1 t^*)^\dagger \Sigma_3 (\Sigma_1 t^*) = t^\dagger \Sigma_1 \Sigma_3 \Sigma_1 t^* = -(t^\dagger \Sigma_3 t)^* = -1.$$

The vector  $\Sigma_1 t^*$  is therefore a correctly normalized hole column of  $T_k$  with signed energy  $-\varepsilon$ , which is  $-\varepsilon_{n,-k}$  for the band of  $t$ . Building the hole columns of  $T_k$  in this way from the particle columns of  $T_{-k}$ , and doing the same with  $k$  and  $-k$  exchanged, yields Eq. (10). The resulting columns are  $\Sigma_3$ -orthogonal to the particle columns of  $T_k$  by Eq. (8), since their signed energies have the opposite sign. A transformation computed independently at  $k$ , such as Colpa's, has hole columns that agree with these up to the freedom of the previous section. At a momentum with  $-k \equiv k$  modulo a reciprocal-lattice vector, the relation constrains  $T_k$  itself and fixes part of that freedom.

## References

### Internal Documents

- *Paraunitary Diagonalization*: states the paraunitary condition, the particle–hole structure of the spectrum, Colpa’s construction, and the gauge freedom proved here.
- *Momentum-Space BdG Hamiltonian*: defines the BdG blocks and the block relations behind the particle–hole relation.
- *Notation and Conventions*: defines the Nambu metric and the paraunitary-matrix notation.
- *Worked Example*: evaluates these statements for a single sublattice.

### External Sources

- J. H. P. Colpa, “Diagonalization of the Quadratic Boson Hamiltonian,” *Physica A* **93**, 327–353 (1978), [doi:10.1016/0378-4371\(78\)90160-7](https://doi.org/10.1016/0378-4371(78)90160-7): metric-preserving diagonalization of a positive-definite quadratic boson Hamiltonian through the Cholesky factorization.

# Worked Example

Source file: docs/lswt/03-examples/worked-example.md — status: draft — updated: 2026-10-01

A single magnetic sublattice,  $N_{\text{sub}} = 1$ , gives the smallest quadratic boson Hamiltonian that contains every ingredient of the general LSWT problem: a normal coefficient, an anomalous pairing coefficient, a paraunitary Bogoliubov transformation, a zero-point correction, and a quantum reduction of the boson number. Every quantity can be written in closed form. The example follows the steps of *Paraunitary Diagonalization* and *Magnon Observables* for this case, so the general formulas can be checked against explicit expressions. We allow the normal coefficients at  $\mathbf{k}$  and  $-\mathbf{k}$  to differ, which shows how a nonreciprocal spectrum  $\varepsilon_{\mathbf{k}} \neq \varepsilon_{-\mathbf{k}}$  arises.

## Single-Sublattice BdG Hamiltonian

For  $N_{\text{sub}} = 1$  the blocks  $A_{\mathbf{k}}$  and  $B_{\mathbf{k}}$  of *Momentum-Space BdG Hamiltonian* are numbers. The block relations reduce to a real normal coefficient  $A_{\mathbf{k}}$  and an even anomalous coefficient,  $B_{-\mathbf{k}} = B_{\mathbf{k}}$ . We drop the sublattice index and write the quadratic Hamiltonian in normal-ordered form,

$$\hat{H}_2 = \sum_{\mathbf{k} \in \text{MBZ}} \left[ A_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \frac{1}{2} (B_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{-\mathbf{k}}^\dagger + B_{\mathbf{k}}^* \hat{a}_{-\mathbf{k}} \hat{a}_{\mathbf{k}}) \right].$$

Symmetrizing the normal term over  $\mathbf{k}$  and  $-\mathbf{k}$  produces the commutator  $\hat{a}_{-\mathbf{k}} \hat{a}_{-\mathbf{k}}^\dagger = \hat{a}_{-\mathbf{k}}^\dagger \hat{a}_{-\mathbf{k}} + 1$ , and the Nambu form therefore carries the constant of the general BdG expression:

$$\hat{H}_2 = \frac{1}{2} \sum_{\mathbf{k} \in \text{MBZ}} \left[ \begin{pmatrix} \hat{a}_{\mathbf{k}}^\dagger & \hat{a}_{-\mathbf{k}} \end{pmatrix} \mathbf{H}_{\mathbf{k}} \begin{pmatrix} \hat{a}_{\mathbf{k}} \\ \hat{a}_{-\mathbf{k}}^\dagger \end{pmatrix} - A_{\mathbf{k}} \right], \quad \mathbf{H}_{\mathbf{k}} = \begin{pmatrix} A_{\mathbf{k}} & B_{\mathbf{k}} \\ B_{\mathbf{k}}^* & A_{-\mathbf{k}} \end{pmatrix}.$$

The relabeling of  $-\mathbf{k}$  to  $\mathbf{k}$  in this step uses that the MBZ momenta form a set closed under inversion. The momentum  $\mathbf{k}$  and its partner  $-\mathbf{k}$  enter only through the even and odd parts of the normal coefficient,

$$A_{\mathbf{k}}^+ = \frac{1}{2} (A_{\mathbf{k}} + A_{-\mathbf{k}}), \quad A_{\mathbf{k}}^- = \frac{1}{2} (A_{\mathbf{k}} - A_{-\mathbf{k}}), \quad \mathbf{H}_{\mathbf{k}} = A_{\mathbf{k}}^- \Sigma_3 + \begin{pmatrix} A_{\mathbf{k}}^+ & B_{\mathbf{k}} \\ B_{\mathbf{k}}^* & A_{\mathbf{k}}^+ \end{pmatrix},$$

where  $\Sigma_3 = \text{diag}(1, -1)$  is the Nambu metric. The odd part  $A_{\mathbf{k}}^-$  is nonzero only when no symmetry of the model and the reference configuration maps  $\mathbf{k}$  to  $-\mathbf{k}$ , for example when a Dzyaloshinskii–Moriya interaction has a component along the ordered moment of a ferromagnet.

## Stability Condition and Magnon Energies

The BdG matrix is positive definite when both diagonal entries and the determinant are positive:

$$A_k > 0, \quad A_{-k} > 0, \quad A_k A_{-k} > |B_k|^2. \quad (11)$$

For  $A_k^- = 0$  these conditions reduce to  $A_k > |B_k|$ . The weaker statement  $|A_k| \geq |B_k|$  is not sufficient: for  $A_k < -|B_k|$  the matrix is negative definite and the magnon vacuum is the highest, not the lowest, state of the harmonic Hamiltonian, and equality gives a zero mode, which is treated in the last section.

The magnon energies are the eigenvalues of  $\Sigma_3 H_k$ . The term  $A_k^- \Sigma_3$  contributes  $A_k^-$  times the identity to  $\Sigma_3 H_k$ , so it shifts both eigenvalues equally, and the remaining matrix has the eigenvalues  $\pm \omega_k$  with

$$\omega_k = \sqrt{(A_k^+)^2 - |B_k|^2}, \quad \varepsilon_k = \omega_k + A_k^-, \quad \varepsilon_{-k} = \omega_k - A_k^-. \quad (12)$$

The eigenvalues of  $\Sigma_3 H_k$  are  $\varepsilon_k$  and  $-\varepsilon_{-k}$ , which is the particle-hole structure of *Paraunitary Diagonalization* for one band. Since  $\omega_{-k} = \omega_k$  and  $A_{-k}^- = -A_k^-$ , the expression for  $\varepsilon_{-k}$  is the same formula evaluated at  $-k$ . The odd part of the normal coefficient thus makes the spectrum nonreciprocal without changing  $\omega_k$ . Under Eq. (11) both energies are positive, because  $\omega_k^2 - (A_k^-)^2 = A_k A_{-k} - |B_k|^2 > 0$ .

The eigenvalues remain real when  $A_k^+ > |B_k|$  but  $A_k A_{-k} < |B_k|^2$ . In that case one of  $\varepsilon_{\pm k}$  is negative,  $H_k$  is indefinite, and creating that magnon lowers the energy, so the magnon vacuum is not the ground state of  $\hat{H}_2$ . Real eigenvalues of  $\Sigma_3 H_k$  therefore do not by themselves establish that the reference configuration is a local energy minimum; positive definiteness of  $H_k$  does.

## Bogoliubov Transformation

We write the anomalous coefficient as  $B_k = |B_k| \exp(i\phi_k)$ , with  $\phi_{-k} = \phi_k$ , and define the Bogoliubov angle  $\theta_k \geq 0$  by

$$\cosh 2\theta_k = \frac{A_k^+}{\omega_k}, \quad \sinh 2\theta_k = \frac{|B_k|}{\omega_k}, \quad \tanh 2\theta_k = \frac{|B_k|}{A_k^+}.$$

The angle is even in  $k$  because  $A_k^+$  and  $|B_k|$  are even. The transformation  $\hat{\Psi}_k = T_k \hat{\Phi}_k$  to the magnon operators  $\hat{b}_k$  is

$$\begin{pmatrix} \hat{a}_k \\ \hat{a}_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} \cosh \theta_k & -\exp(i\phi_k) \sinh \theta_k \\ -\exp(-i\phi_k) \sinh \theta_k & \cosh \theta_k \end{pmatrix} \begin{pmatrix} \hat{b}_k \\ \hat{b}_{-k}^\dagger \end{pmatrix}. \quad (13)$$

In the notation of the general transformation,  $P_k = \cosh \theta_k$  and  $Q_k = -\exp(i\phi_k) \sinh \theta_k$ ; the lower row is the Hermitian conjugate of the upper row at  $-k$  because  $\theta_k$  and  $\phi_k$  are even. The paraunitary condition  $T_k \Sigma_3 T_k^\dagger = \Sigma_3$  reduces to  $\cosh^2 \theta_k - \sinh^2 \theta_k = 1$  on the diagonal, and its off-diagonal entries cancel. The commutator  $[\hat{a}_k, \hat{a}_k^\dagger] = 1$  therefore holds when it holds for the magnon operators.

The phase  $\phi_k$  can be removed by redefining  $\hat{a}_k \rightarrow \exp(i\phi_k/2) \hat{a}_k$ , which makes  $B_k$  real and nonnegative. Keeping the phase explicit shows where it appears in the anomalous average below.

## Diagonal Hamiltonian

Only the second matrix of the decomposition of  $H_k$  needs to be diagonalized, since  $T_k^\dagger \Sigma_3 T_k = \Sigma_3$  leaves the term  $A_k^- \Sigma_3$  unchanged. For that matrix the transformation gives

$$T_k^\dagger \begin{pmatrix} A_k^+ & B_k \\ B_k^* & A_k^+ \end{pmatrix} T_k = \begin{pmatrix} A_k^+ \cosh 2\theta_k - |B_k| \sinh 2\theta_k & \exp(i\phi_k)(|B_k| \cosh 2\theta_k - A_k^+ \sinh 2\theta_k) \\ \exp(-i\phi_k)(|B_k| \cosh 2\theta_k - A_k^+ \sinh 2\theta_k) & A_k^+ \cosh 2\theta_k - |B_k| \sinh 2\theta_k \end{pmatrix}.$$

The off-diagonal entries vanish for the angle  $\tanh 2\theta_k = |B_k|/A_k^+$ , and the diagonal entries equal  $[(A_k^+)^2 - |B_k|^2]/\omega_k = \omega_k$ . Adding back  $A_k^- \Sigma_3$  gives  $T_k^\dagger H_k T_k = \text{diag}(\varepsilon_k, \varepsilon_{-k})$ , in agreement with Eq. (12). The diagonal magnon Hamiltonian and its zero-point correction follow from *Paraunitary Diagonalization*:

$$\hat{H}_2 = \sum_{k \in \text{MBZ}} \varepsilon_k \hat{b}_k^\dagger \hat{b}_k + \Delta E_{\text{zp}}, \quad \Delta E_{\text{zp}} = \frac{1}{2} \sum_{k \in \text{MBZ}} (\varepsilon_k - A_k) = \frac{1}{2} \sum_{k \in \text{MBZ}} (\omega_k - A_k^+). \quad (14)$$

The second form follows because  $A_k^-$  is odd and cancels in the sum over a set closed under inversion. Each summand  $\omega_k - A_k^+$  is negative when  $B_k \neq 0$ , so the anomalous terms lower the energy below the classical value; the nonreciprocal part does not change the zero-point correction. For  $B_k = 0$  the angle vanishes,  $T_k$  is the identity, and  $\Delta E_{\text{zp}} = 0$ : the magnon vacuum coincides with the boson vacuum.

## Boson Number and Anomalous Average

Inserting Eq. (13) into the averages and using the Gibbs occupations  $n_k = n_B(\varepsilon_k)$  of *Thermodynamics*, with  $\langle \hat{b}_k^\dagger \hat{b}_k \rangle = n_k$  and vanishing averages of  $\hat{b}\hat{b}$  and of products at different momenta, gives

$$\begin{aligned} \langle \hat{a}_k^\dagger \hat{a}_k \rangle &= \cosh^2 \theta_k n_k + \sinh^2 \theta_k (1 + n_{-k}), \\ \langle \hat{a}_k \hat{a}_{-k} \rangle &= -\exp(i\phi_k) \cosh \theta_k \sinh \theta_k (1 + n_k + n_{-k}). \end{aligned} \quad (15)$$

The first line is the single-band form of the boson number in *Magnon Observables*. In the magnon vacuum the occupations vanish, and

$$\langle \hat{a}_k^\dagger \hat{a}_k \rangle_0 = \sinh^2 \theta_k = \frac{1}{2} \left( \frac{A_k^+}{\omega_k} - 1 \right), \quad \langle \hat{a}_k \hat{a}_{-k} \rangle_0 = -\frac{B_k}{2\omega_k}.$$

The zero-temperature boson number is the quantum reduction of the ordered moment. It is nonzero only when  $B_k \neq 0$ , that is, when the harmonic Hamiltonian does not conserve the number of Holstein–Primakoff bosons. The anomalous average carries the phase of  $B_k$  and is the off-diagonal block of the correlation matrix  $T_k N_k(0) T_k^\dagger$ .

## Response to a Uniform Shift of the Normal Coefficient

The zero-point correction and the boson number are related by the Hellmann–Feynman theorem. Adding  $\mu \sum_k \hat{a}_k^\dagger \hat{a}_k$  to  $\hat{H}_2$  shifts  $A_k \rightarrow A_k + \mu$  and leaves  $B_k$  unchanged. The derivative of the ground-state energy with respect to  $\mu$  is the ground-state average of the added operator, and Eq. (14) gives

$$\frac{\partial \Delta E_{zp}}{\partial \mu} = \frac{1}{2} \sum_{\mathbf{k} \in \text{MBZ}} \left( \frac{A_{\mathbf{k}}^+ + \mu}{\omega_{\mathbf{k}}(\mu)} - 1 \right) = \sum_{\mathbf{k} \in \text{MBZ}} \langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \rangle_0, \quad \omega_{\mathbf{k}}(\mu) = \sqrt{(A_{\mathbf{k}}^+ + \mu)^2 - |B_{\mathbf{k}}|^2}.$$

This identity provides an independent check of the zero-temperature boson number: the boson number can be obtained either from the transformation  $T_{\mathbf{k}}$  or from the energies alone. To second order in  $\mu$ ,

$$\Delta E_{zp}(\mu) = \Delta E_{zp}(0) + \mu \sum_{\mathbf{k}} \langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \rangle_0 - \frac{\mu^2}{4} \sum_{\mathbf{k}} \frac{|B_{\mathbf{k}}|^2}{\omega_{\mathbf{k}}^3} + O(\mu^3), \quad \frac{\partial \langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \rangle_0}{\partial \mu} = -\frac{|B_{\mathbf{k}}|^2}{2\omega_{\mathbf{k}}^3},$$

with  $\omega_{\mathbf{k}} = \omega_{\mathbf{k}}(0)$ . The second-order coefficient is negative, as second-order perturbation theory requires for the ground-state energy, and the boson number decreases when the normal coefficient grows relative to the pairing.

## Approach to a Zero Mode

The stability condition becomes marginal when  $A_{\mathbf{k}} A_{-\mathbf{k}} \rightarrow |B_{\mathbf{k}}|^2$ . For  $A_{\mathbf{k}}^- = 0$  this is the limit  $\omega_{\mathbf{k}} \rightarrow 0^+$  at fixed  $A_{\mathbf{k}}^+ > 0$ , which is the single-mode Goldstone case  $A = B > 0$  of *Paraunitary Diagonalization*. In this limit  $\cosh \theta_{\mathbf{k}}$  and  $\sinh \theta_{\mathbf{k}}$  grow as  $(A_{\mathbf{k}}^+ / 2\omega_{\mathbf{k}})^{1/2}$ , and

$$\langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \rangle_0 \simeq \frac{A_{\mathbf{k}}^+}{2\omega_{\mathbf{k}}}, \quad \langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \rangle \simeq \frac{A_{\mathbf{k}}^+}{\omega_{\mathbf{k}}} n_{\text{B}}(\omega_{\mathbf{k}}) \simeq \frac{A_{\mathbf{k}}^+ k_{\text{B}} T}{\omega_{\mathbf{k}}^2} \quad (\omega_{\mathbf{k}} \ll k_{\text{B}} T).$$

The occupation per mode diverges at the zero mode, while the energy per mode,  $\frac{1}{2}(\omega_{\mathbf{k}} - A_{\mathbf{k}}^+)$ , stays finite. Whether the momentum sum of the boson number converges then depends on the dispersion near the zero and on the dimension, as discussed in *Magnon Observables*. At  $\omega_{\mathbf{k}} = 0$  itself the transformation does not exist, and the single-mode boson number is undefined rather than large.

## References

### Internal Documents

- *Momentum-Space BdG Hamiltonian*: defines the Nambu spinor, the BdG blocks, their block relations, and the trace subtraction specialized here.
- *Paraunitary Diagonalization*: supplies the paraunitary condition, the particle–hole structure, the zero-point correction, and the single-mode zero-mode cases.
- *Magnon Observables*: defines the correlation matrix and the boson number whose single-band forms are evaluated here.
- *Thermodynamics*: defines the Bose–Einstein occupations.
- *Notation and Conventions*: defines  $\Sigma_3$ ,  $T_{\mathbf{k}}$ ,  $\hat{\Phi}_{\mathbf{k}}$ , and the energy symbols.

### External Sources

- None.

# Magnon Observables

Source file: docs/lswt/02-observables/magnon-observables.md — status: draft — updated: 2026-10-01

After the paraunitary diagonalization of *Paraunitary Diagonalization*, every LSWT observable is built from two ingredients at each momentum of the magnetic Brillouin zone (MBZ): the magnon energies  $\varepsilon_{nk}$  and the transformation  $T_k$ . This document collects the quantities that follow directly from these ingredients, namely the magnon spectrum, the equal-time correlation matrix of the Holstein–Primakoff bosons, the boson number, and the reduced ordered moment. It also serves as an index to the documents that own the remaining observables.

## Magnon Spectrum and Band Structure

The magnon energies  $\varepsilon_{nk}$ ,  $n = 1, \dots, N_{\text{sub}}$ , are the positive eigenvalues of  $\Sigma_3 H_k$  and are periodic over the MBZ. Each band is a branch  $\varepsilon_{nk}$  that continues across momentum; at crossings the band label is a matter of convention, since the energies are ordered at each momentum. The negative eigenvalues  $-\varepsilon_{n,-k}$  are the particle–hole partners and carry no additional information.

The magnetic unit cell of an ordered state may contain several crystallographic unit cells. A band structure plotted along a path of the crystallographic Brillouin zone (CBZ) then shows all  $N_{\text{sub}}$  bands of the magnetic cell at every momentum, including branches folded back by magnetic reciprocal-lattice vectors. Which folded branch carries weight at a given external momentum is determined by the one-magnon weights of *Structure Factor and Spectral Function*, not by the energies alone. At a momentum where  $H_k$  is only positive semidefinite, some  $\varepsilon_{nk}$  vanishes; the energies remain well defined as limits, although  $T_k$  may not exist there.

## Ground-State Energy

The LSWT ground-state energy  $E_{\text{GS}} = E_{\text{cl}} + \Delta E_{\text{zp}}$  and the zero-point correction  $\Delta E_{\text{zp}} = \frac{1}{2} \sum_k (\sum_n \varepsilon_{nk} - \text{Tr } A_k)$  are derived with the diagonal magnon Hamiltonian in *Paraunitary Diagonalization*. The energy per magnetic site is  $\mathcal{E}_{\text{GS}} = E_{\text{GS}}/N_{\text{site}}$ . The correction depends on the energies alone, not on  $T_k$ , and it inherits the provisional trace convention of *Momentum-Space BdG Hamiltonian*.

## Equal-Time Correlation Matrix

The equal-time averages of all products of two Holstein–Primakoff bosons at momentum  $k$  form the  $2N_{\text{sub}} \times 2N_{\text{sub}}$  correlation matrix

$$[\langle \hat{\Psi}_k \hat{\Psi}_k^\dagger \rangle] = \begin{pmatrix} \langle \hat{a}_{k\mu} \hat{a}_{k\nu}^\dagger \rangle & \langle \hat{a}_{k\mu} \hat{a}_{-k\nu} \rangle \\ \langle \hat{a}_{-k\mu}^\dagger \hat{a}_{k\nu} \rangle & \langle \hat{a}_{-k\mu}^\dagger \hat{a}_{-k\nu} \rangle \end{pmatrix}_{\mu\nu} = \mathbf{T}_k \mathbf{N}_k(0) \mathbf{T}_k^\dagger, \quad (16)$$

where each block is an  $N_{\text{sub}} \times N_{\text{sub}}$  matrix with the sublattice indices  $\mu, \nu$ . The second equality substitutes  $\hat{\Psi}_k = \mathbf{T}_k \hat{\Phi}_k$ . In the Gibbs state the magnon correlation matrix  $\mathbf{N}_k(0) = \text{diag}(1 + n_{1k}, \dots, n_{1,-k}, \dots)$  of *Spin Correlations* is diagonal, with the occupations  $n_{nk} = n_B(\varepsilon_{nk})$  of *Thermodynamics*. In the ground state, the magnon vacuum, the occupations vanish and  $\mathbf{N}_k(0) = \text{diag}(\mathbf{I}_{N_{\text{sub}}}, 0)$ . The anomalous averages  $\langle \hat{a}\hat{a} \rangle$  are nonzero even at zero temperature when  $H_k$  contains the anomalous block  $B_k$ .

## Boson Number and Reduced Moment

The average number of Holstein–Primakoff bosons on sublattice  $\mu$  is  $\langle \hat{n}_\mu \rangle = N_{\text{uc}}^{-1} \sum_i \langle \hat{a}_{i\mu}^\dagger \hat{a}_{i\mu} \rangle = N_{\text{uc}}^{-1} \sum_k \langle \hat{a}_{k\mu}^\dagger \hat{a}_{k\mu} \rangle$ . The lower-right block of Eq. (16) contains these averages at  $-k$ , and the momentum sum covers all momenta, so

$$\langle \hat{n}_\mu \rangle = \frac{1}{N_{\text{uc}}} \sum_{k \in \text{MBZ}} [\mathbf{T}_k \mathbf{N}_k(0) \mathbf{T}_k^\dagger]_{N_{\text{sub}} + \mu, N_{\text{sub}} + \mu} = \frac{1}{N_{\text{uc}}} \sum_{k \in \text{MBZ}} \sum_{n=1}^{N_{\text{sub}}} [|\langle P_k \rangle_{\mu n}|^2 n_{nk} + |\langle Q_k \rangle_{\mu n}|^2 (1 + n_{nk})]. \quad (17)$$

The second form uses the blocks of the Bogoliubov transformation  $\hat{a}_{k\mu} = \sum_n [(\langle P_k \rangle_{\mu n} \hat{b}_{kn} + (\langle Q_{-k} \rangle_{\mu n} \hat{b}_{-kn}^\dagger)]$  of *Paraunitary Diagonalization* and relabels  $-k \rightarrow k$  in the hole-column terms. The  $Q$  term survives at zero temperature: the quantum reduction  $N_{\text{uc}}^{-1} \sum_{k,n} |\langle Q_k \rangle_{\mu n}|^2$  arises only from the mixing of creation and annihilation operators and vanishes when  $B_k = 0$ , as for a collinear ferromagnet whose Hamiltonian conserves the total spin component along the ordered moment. Exchange anisotropy transverse to the moment, such as  $J^{xx} \neq J^{yy}$  in the local frame or Kitaev and off-diagonal couplings, produces  $B_k \neq 0$  and a quantum reduction even in a collinear ferromagnet. The  $P$  and  $Q$  terms proportional to  $n_{nk}$  give the thermal reduction.

The boson number reduces the ordered moment. With  $\hat{S}_I^0 = S_I - \hat{n}_I$  and the local longitudinal direction  $\mathbf{n}_\mu$  of *Classical Order and Local Frame*, the average spin of sublattice  $\mu$  is

$$\mathbf{m}_\mu = (S_\mu - \langle \hat{n}_\mu \rangle) \mathbf{n}_\mu, \quad (18)$$

since the transverse components average to zero. The magnetization per site is  $N_{\text{sub}}^{-1} \sum_\mu \mathbf{m}_\mu$ . These moments enter the elastic Bragg scattering of *Spin Correlations*. The reduction  $\langle \hat{n}_\mu \rangle$  is of order  $S^0$ , compared with  $S_\mu$ , so the expansion is controlled only while  $\langle \hat{n}_\mu \rangle \ll S_\mu$ . A value  $\langle \hat{n}_\mu \rangle \geq S_\mu$  would remove or reverse the moment and signals that LSWT about the chosen reference configuration no longer applies.

## Zero Modes

At a zero mode the summand of Eq. (17) can diverge. Near a Goldstone mode of an antiferromagnet,  $\varepsilon_{nk} \propto |k - k_0|$  and  $|\langle P_k \rangle|^2, |\langle Q_k \rangle|^2 \propto 1/\varepsilon_{nk}$ , as for the single mode of *Paraunitary Diagonalization*. At zero temperature the summand then grows as  $1/|k - k_0|$ , which is integrable in two dimensions, and the quantum reduction is finite. At nonzero temperature  $n_B(\varepsilon) \simeq k_B T / \varepsilon$  adds a factor  $1/\varepsilon$ , and the

summand grows as  $|k - k_0|^{-2}$ ; the same power follows for a ferromagnetic Goldstone mode with  $\varepsilon \propto |k - k_0|^2$  and bounded  $P_k$ . In two dimensions this gives a logarithmically divergent thermal boson number. The divergence is the LSWT form of the absence of spontaneous continuous-symmetry breaking at nonzero temperature in two dimensions, discussed in *LSWT Overview*. The thermodynamic potentials of *Thermodynamics* remain finite in the same situation.

## Index of Observables

| Observable                                               | Ingredients                            | Owner document                                    |
|----------------------------------------------------------|----------------------------------------|---------------------------------------------------|
| Magnon energies and bands                                | $\varepsilon_{nk}$                     | This document; <i>Paraunitary Diagonalization</i> |
| Ground-state energy, zero-point correction               | $\varepsilon_{nk}, \text{Tr } A_k$     | <i>Paraunitary Diagonalization</i>                |
| Boson number, reduced moment, magnetization              | $T_k, n_{nk}$                          | This document                                     |
| Partition function, $F, U, \mathcal{S}, C$               | $\varepsilon_{nk}$                     | <i>Thermodynamics</i>                             |
| Spin correlation function, elastic correlation           | $T_q, N_q(t), m_\mu$                   | <i>Spin Correlations</i>                          |
| Static and dynamic structure factors, spectral function  | One-magnon weights $W_n^{\alpha\beta}$ | <i>Structure Factor and Spectral Function</i>     |
| Berry curvature, Chern number, thermal Hall conductivity | $T_k, \partial_k H_k$                  | <i>Topological Magnon Quantities</i>              |
| Classical lower bound and candidate orders               | Classical energy only                  | <i>Appendix: Luttinger–Tisza Method</i>           |

## References

### Internal Documents

- *Paraunitary Diagonalization*: supplies  $\varepsilon_{nk}, T_k$  with its blocks  $P_k$  and  $Q_k$ , and the ground-state energy.
- *Momentum-Space BdG Hamiltonian*: defines the Nambu spinor and the BdG blocks.
- *Classical Order and Local Frame*: defines the ordered directions  $n_\mu$ .
- *Thermodynamics*: defines the occupations and the thermodynamic potentials.
- *Spin Correlations*: defines  $N_q(t)$  and uses the reduced moments.
- *Structure Factor and Spectral Function*: determines the spectral weight of folded bands.
- *Topological Magnon Quantities*: builds topological responses from the bands and eigenvectors.
- *Appendix: Luttinger–Tisza Method*: bounds the classical energy of candidate reference configurations.
- *LSWT Overview*: states the finite-temperature qualification in two dimensions.

### External Sources

- None.

# Thermodynamics

Source file: docs/lswt/02-observables/thermodynamics.md — status: draft — updated: 2026-09-30

In thermal equilibrium the LSWT Hamiltonian describes an ideal gas of magnons on top of the classical reference configuration. The partition function therefore factorizes over the magnon modes  $(\mathbf{k}, n)$ , and the free energy, internal energy, entropy, and heat capacity follow from single-oscillator expressions summed over the magnetic Brillouin zone (MBZ). Energy, temperature, and entropy symbols follow *Notation and Conventions*; the inverse temperature is  $\beta = (k_B T)^{-1}$  and the entropy is  $\mathcal{S}$ .

## Validity of the Magnon-Gas Description

We use the diagonal Hamiltonian of *Paraunitary Diagonalization*,

$$\hat{H}_{\text{LSWT}} = E_{\text{cl}} + \hat{H}_2 = E_{\text{GS}} + \sum_{\mathbf{k} \in \text{MBZ}} \sum_{n=1}^{N_{\text{sub}}} \varepsilon_{n\mathbf{k}} \hat{n}_{\mathbf{k}n}, \quad \hat{n}_{\mathbf{k}n} = \hat{b}_{\mathbf{k}n}^\dagger \hat{b}_{\mathbf{k}n},$$

with  $E_{\text{GS}} = E_{\text{cl}} + \Delta E_{\text{zp}}$ . The classical reference configuration and the magnon energies  $\varepsilon_{n\mathbf{k}}$  are held fixed as the temperature changes, and the quartic and higher Holstein–Primakoff terms that couple magnons are omitted. The results below are therefore the leading low-temperature description. Its corrections grow with the thermal magnon density, and the description fails when the thermal reduction of a sublattice moment, discussed in *Magnon Observables*, becomes comparable to the spin length. In two dimensions, the interpretation of these results as properties of an ordered phase requires the qualifications stated in *LSWT Overview*.

All quantities below assume  $\varepsilon_{n\mathbf{k}} > 0$ , which holds when  $H_{\mathbf{k}}$  is positive definite. The section on zero modes describes the limit  $\varepsilon_{n\mathbf{k}} \rightarrow 0$ .

## Partition Function

The Gibbs state  $\hat{\rho} = \exp(-\beta \hat{H}_{\text{LSWT}})/Z$  has the partition function  $Z = \text{Tr} \exp(-\beta \hat{H}_{\text{LSWT}})$ . The number operators of different modes commute, so the trace factorizes into independent geometric series, one for each mode:

$$Z = \exp(-\beta E_{\text{GS}}) \prod_{\mathbf{k} \in \text{MBZ}} \prod_{n=1}^{N_{\text{sub}}} \frac{1}{1 - \exp(-\beta \varepsilon_{n\mathbf{k}})}, \quad -\ln Z = \beta E_{\text{GS}} + \sum_{\mathbf{k} \in \text{MBZ}} \sum_{n=1}^{N_{\text{sub}}} \ln(1 - \exp(-\beta \varepsilon_{n\mathbf{k}})). \quad (19)$$

Each geometric series converges because  $\varepsilon_{nk} > 0$ . The constant  $E_{\text{GS}}$  contributes the factor  $\exp(-\beta E_{\text{GS}})$ ; it shifts the internal and free energies but drops out of the entropy and heat capacity, which depend only on the thermal part of  $\ln Z$ .

The thermal occupation of mode  $(k, n)$  is the Bose–Einstein distribution

$$\langle \hat{n}_{kn} \rangle = n_{\text{B}}(\varepsilon_{nk}), \quad n_{\text{B}}(\varepsilon) = \frac{1}{\exp(\beta\varepsilon) - 1}. \quad (20)$$

## Internal and Free Energies

The internal energy is the thermal average of the Hamiltonian,  $U = \langle \hat{H}_{\text{LSWT}} \rangle = -\partial \ln Z / \partial \beta$ . Differentiating Eq. (19) gives

$$U = E_{\text{GS}} + \sum_{k \in \text{MBZ}} \sum_{n=1}^{N_{\text{sub}}} \varepsilon_{nk} n_{\text{B}}(\varepsilon_{nk}). \quad (21)$$

The internal energy thus consists of the classical energy, the zero-point correction, and the energy of the thermally excited magnons. The Helmholtz free energy, defined by  $Z = \exp(-\beta F)$ , is

$$F = E_{\text{GS}} + k_{\text{B}} T \sum_{k \in \text{MBZ}} \sum_{n=1}^{N_{\text{sub}}} \ln(1 - \exp(-\beta \varepsilon_{nk})). \quad (22)$$

Each logarithm is negative, so the thermal magnons lower  $F$  below  $E_{\text{GS}}$ . At  $T = 0$  both  $U$  and  $F$  reduce to  $E_{\text{GS}}$ .

## Entropy

The entropy of the Gibbs state is the von Neumann entropy  $\mathcal{S} = -k_{\text{B}} \text{Tr}(\hat{\rho} \ln \hat{\rho})$ . Inserting  $\ln \hat{\rho} = -\beta \hat{H}_{\text{LSWT}} - \ln Z$  gives  $\mathcal{S} = (U - F)/T$ , which coincides with the thermodynamic relation  $\mathcal{S} = -\partial F / \partial T$ . With Eq. (21) and Eq. (22),

$$\mathcal{S} = k_{\text{B}} \sum_{k \in \text{MBZ}} \sum_{n=1}^{N_{\text{sub}}} [\beta \varepsilon_{nk} n_{\text{B}}(\varepsilon_{nk}) - \ln(1 - \exp(-\beta \varepsilon_{nk}))].$$

The relations  $\beta \varepsilon = \ln[(1 + n_{\text{B}})/n_{\text{B}}]$  and  $\ln(1 - \exp(-\beta \varepsilon)) = -\ln(1 + n_{\text{B}})$ , which follow from the definition of  $n_{\text{B}}$ , express the entropy through the occupations alone:

$$\mathcal{S} = k_{\text{B}} \sum_{k \in \text{MBZ}} \sum_{n=1}^{N_{\text{sub}}} [(1 + n_{kn}) \ln(1 + n_{kn}) - n_{kn} \ln n_{kn}], \quad n_{kn} = n_{\text{B}}(\varepsilon_{nk}). \quad (23)$$

This is the entropy of an ideal Bose gas with the occupations  $n_{kn}$ . The zero-point correction does not enter, since it is a temperature-independent constant.

## Heat Capacity

The heat capacity is the temperature derivative of the internal energy,  $C = \partial U / \partial T$ . Only the occupations depend on temperature, and  $\partial n_B(\varepsilon) / \partial T = (\varepsilon / k_B T^2) \exp(\beta \varepsilon) / (\exp(\beta \varepsilon) - 1)^2$ , so

$$C = k_B \sum_{\mathbf{k} \in \text{MBZ}} \sum_{n=1}^{N_{\text{sub}}} \frac{(\beta \varepsilon_{n\mathbf{k}})^2}{4 \sinh^2(\beta \varepsilon_{n\mathbf{k}}/2)}. \quad (24)$$

Each mode contributes the heat capacity of a harmonic oscillator. The contribution approaches  $k_B$  for  $\varepsilon_{n\mathbf{k}} \ll k_B T$  and is exponentially small for  $\varepsilon_{n\mathbf{k}} \gg k_B T$ , so at low temperature the heat capacity is dominated by the lowest magnon energies. The relation  $C = T \partial \mathcal{S} / \partial T$  gives the same expression.

## Normalization and Momentum Sums

The quantities above are extensive: the sums run over the  $N_{\text{uc}}$  momenta of the MBZ and the  $N_{\text{sub}}$  bands. Values per magnetic site follow by dividing by  $N_{\text{site}} = N_{\text{uc}} N_{\text{sub}}$ ; for example,  $U / N_{\text{site}} = \mathcal{E}_{\text{GS}} + N_{\text{site}}^{-1} \sum_{\mathbf{k}, n} \varepsilon_{n\mathbf{k}} n_B(\varepsilon_{n\mathbf{k}})$ . In the thermodynamic limit the momentum sum becomes an integral over the MBZ,  $N_{\text{uc}}^{-1} \sum_{\mathbf{k}} \rightarrow A_{\text{MBZ}}^{-1} \int_{\text{MBZ}} d^2 \mathbf{k}$ , where  $A_{\text{MBZ}}$  is the area of the MBZ. A finite momentum mesh approximates this integral.

## Zero Modes

When  $H_{\mathbf{k}}$  is only positive semidefinite, some magnon energy vanishes, for example at a Goldstone mode. Near an isolated zero at  $\mathbf{k}_0$  we assume  $\varepsilon_{n\mathbf{k}} \propto |\mathbf{k} - \mathbf{k}_0|^a$  with  $a > 0$ ; linear ( $a = 1$ ) and quadratic ( $a = 2$ ) dispersions are the common cases. The summands behave as follows as  $\varepsilon \rightarrow 0$ :

| Quantity      | Summand for $\varepsilon \ll k_B T$              | Limit at $\varepsilon = 0$ |
|---------------|--------------------------------------------------|----------------------------|
| $U$           | $\varepsilon n_B(\varepsilon) \rightarrow k_B T$ | finite                     |
| $C$           | $k_B$                                            | finite                     |
| $F$           | $k_B T \ln(\beta \varepsilon)$                   | logarithmically divergent  |
| $\mathcal{S}$ | $k_B [1 + \ln(k_B T / \varepsilon)]$             | logarithmically divergent  |

The logarithmic divergences are integrable. In two dimensions  $\int d^2 \mathbf{q} \ln |\mathbf{q}|$  converges near  $\mathbf{q} = 0$  for any  $a > 0$ , so  $F$ ,  $U$ ,  $\mathcal{S}$ , and  $C$  remain finite in the thermodynamic limit. A zero mode at an isolated momentum has vanishing weight in the MBZ integral. On a finite mesh that contains  $\mathbf{k}_0$ , the treatment of that single point is a discretization choice, and the continuum values do not depend on it. Quantities weighted by the occupations without a compensating factor of  $\varepsilon$ , such as the boson number and the sublattice moment, can diverge instead; they are treated in *Magnon Observables*.

## References

### Internal Documents

- *Notation and Conventions*: defines  $\beta$ ,  $\mathcal{S}$ ,  $\varepsilon_{n\mathbf{k}}$ ,  $E_{\text{cl}}$ ,  $\Delta E_{\text{zp}}$ ,  $E_{\text{GS}}$ , and the system-size symbols.
- *Paraunitary Diagonalization*: supplies the diagonal magnon Hamiltonian, the zero-point correction, and the positive-definiteness condition.

- *LSWT Overview*: states the finite-temperature qualification for two-dimensional ordered states.
- *Magnon Observables*: uses the Bose–Einstein occupations for boson numbers and sublattice moments.
- *Thermodynamic Derivations*: is reserved for longer entropy and correlation-matrix derivations.

### External Sources

- None.

# Appendix: Thermodynamic Derivations

Source file: docs/lswt/04-appendices/thermodynamic-derivations.md — status: draft — updated: 2026-10-01

The thermodynamic quantities of a harmonic magnon gas and the equal-time correlation matrix of the Holstein–Primakoff bosons both follow from the Gibbs state of the diagonal magnon Hamiltonian. *Thermodynamics* and *Magnon Observables* state the results with short derivations; this appendix gives the intermediate steps: the factorization of the partition function, the entropy and heat-capacity identities, the small-energy expansions behind the zero-mode table, and the correlation-matrix route to the boson number. We assume, as in those documents, that  $H_k$  is positive definite at every momentum of the magnetic Brillouin zone (MBZ), so that every magnon energy  $\varepsilon_{nk}$  is positive, except where the zero-mode limit is discussed.

## Factorization of the Partition Function

The diagonal Hamiltonian  $\hat{H}_{\text{LSWT}} = E_{\text{GS}} + \sum_{k,n} \varepsilon_{nk} \hat{n}_{kn}$  of *Paraunitary Diagonalization* is diagonal in the Fock basis  $|m_{kn}\rangle$  of magnon occupation numbers  $m_{kn} = 0, 1, 2, \dots$ , with eigenvalues  $E_{\text{GS}} + \sum_{k,n} \varepsilon_{nk} m_{kn}$ . The trace is a sum over all occupation configurations, and the exponential of a sum is a product, so

$$Z = \exp(-\beta E_{\text{GS}}) \prod_{k,n} \sum_{m=0}^{\infty} \exp(-\beta \varepsilon_{nk} m) = \exp(-\beta E_{\text{GS}}) \prod_{k,n} \frac{1}{1 - \exp(-\beta \varepsilon_{nk})}.$$

Each geometric series converges because  $\varepsilon_{nk} > 0$ . The Gibbs state is the product of the single-mode thermal states  $\hat{\rho}_{kn} = (1 - \exp(-\beta \varepsilon_{nk})) \exp(-\beta \varepsilon_{nk} \hat{n}_{kn})$ . The occupation of one mode follows from the derivative of the corresponding factor,  $\langle \hat{n}_{kn} \rangle = -\beta^{-1} \partial \ln Z / \partial \varepsilon_{nk} = [\exp(\beta \varepsilon_{nk}) - 1]^{-1} = n_{\text{B}}(\varepsilon_{nk})$ .

## Internal Energy and Free Energy

The internal energy is  $U = \text{Tr}(\hat{\rho} \hat{H}_{\text{LSWT}}) = -\partial \ln Z / \partial \beta$ . Differentiating the logarithm of each factor gives

$$-\frac{\partial}{\partial \beta} \ln \frac{1}{1 - \exp(-\beta \varepsilon)} = \frac{\varepsilon \exp(-\beta \varepsilon)}{1 - \exp(-\beta \varepsilon)} = \varepsilon n_{\text{B}}(\varepsilon),$$

and the prefactor contributes  $E_{\text{GS}}$ , which yields  $U = E_{\text{GS}} + \sum_{k,n} \varepsilon_{nk} n_{\text{B}}(\varepsilon_{nk})$ . The free energy  $F = -\beta^{-1} \ln Z$  follows by taking the logarithm of the product,  $F = E_{\text{GS}} + \beta^{-1} \sum_{k,n} \ln(1 - \exp(-\beta \varepsilon_{nk}))$ .

## Entropy

The von Neumann entropy of the Gibbs state uses  $\ln \hat{\rho} = -\beta \hat{H}_{\text{LSWT}} - \ln Z$ :

$$\mathcal{S} = -k_B \text{Tr}(\hat{\rho} \ln \hat{\rho}) = k_B \beta \text{Tr}(\hat{\rho} \hat{H}_{\text{LSWT}}) + k_B \ln Z = \frac{U - F}{T}.$$

The thermodynamic entropy  $-\partial F/\partial T$  gives the same result. With  $F = -k_B T \ln Z$  and  $\partial \beta/\partial T = -1/(k_B T^2)$ ,

$$-\frac{\partial F}{\partial T} = k_B \ln Z + k_B T \frac{\partial \ln Z}{\partial \beta} \frac{\partial \beta}{\partial T} = k_B \ln Z + \frac{U}{T} = \frac{U - F}{T}.$$

For the Gibbs state the von Neumann entropy therefore reduces to the thermodynamic entropy. Inserting the expressions for  $U$  and  $F$ , the constant  $E_{\text{GS}}$  cancels in  $U - F$ , and

$$\mathcal{S} = k_B \sum_{\mathbf{k}, n} [\beta \varepsilon_{n\mathbf{k}} n_{\mathbf{k}n} - \ln(1 - \exp(-\beta \varepsilon_{n\mathbf{k}}))], \quad n_{\mathbf{k}n} = n_B(\varepsilon_{n\mathbf{k}}).$$

The definition  $n = [\exp(\beta \varepsilon) - 1]^{-1}$  gives  $\exp(\beta \varepsilon) = (1 + n)/n$  and  $1 - \exp(-\beta \varepsilon) = 1/(1 + n)$ , so

$$\beta \varepsilon = \ln \frac{1 + n}{n}, \quad \ln(1 - \exp(-\beta \varepsilon)) = -\ln(1 + n).$$

Substituting these relations, the summand becomes  $n \ln(1 + n) - n \ln n + \ln(1 + n) = (1 + n) \ln(1 + n) - n \ln n$ , which is the occupation form of the entropy in *Thermodynamics*.

## Heat Capacity

The temperature derivative of the occupation is

$$\frac{\partial n_B(\varepsilon)}{\partial T} = \frac{\varepsilon}{k_B T^2} \frac{\exp(\beta \varepsilon)}{(\exp(\beta \varepsilon) - 1)^2} = \frac{\varepsilon}{k_B T^2} \frac{1}{4 \sinh^2(\beta \varepsilon/2)},$$

where the second form uses  $\exp(x)/(\exp(x) - 1)^2 = 1/(\exp(x/2) - \exp(-x/2))^2$ . Since only the occupations depend on temperature,  $C = \partial U/\partial T = \sum_{\mathbf{k}, n} \varepsilon_{n\mathbf{k}} \partial n_{\mathbf{k}n}/\partial T$ , which gives the expression  $k_B \sum (\beta \varepsilon)^2/[4 \sinh^2(\beta \varepsilon/2)]$ .

The relation  $C = T \partial \mathcal{S}/\partial T$  follows from the occupation form of the entropy. The derivative of the summand with respect to the occupation is  $k_B \ln[(1 + n)/n] = k_B \beta \varepsilon$ , so

$$T \frac{\partial \mathcal{S}}{\partial T} = T \sum_{\mathbf{k}, n} k_B \beta \varepsilon_{n\mathbf{k}} \frac{\partial n_{\mathbf{k}n}}{\partial T} = \sum_{\mathbf{k}, n} \varepsilon_{n\mathbf{k}} \frac{\partial n_{\mathbf{k}n}}{\partial T} = \frac{\partial U}{\partial T}.$$

## Small-Energy Expansions Behind the Zero-Mode Limits

With  $x = \beta\varepsilon \rightarrow 0^+$ , the summands of the thermodynamic quantities have the expansions

$$\begin{aligned}\varepsilon n_B(\varepsilon) &= k_B T \left( 1 - \frac{x}{2} + \frac{x^2}{12} + O(x^4) \right), \\ k_B T \ln(1 - \exp(-x)) &= k_B T \left( \ln x - \frac{x}{2} + \frac{x^2}{24} + O(x^4) \right), \\ k_B [x n_B - \ln(1 - \exp(-x))] &= k_B \left( 1 - \ln x + \frac{x^2}{24} + O(x^4) \right), \\ k_B \frac{x^2}{4 \sinh^2(x/2)} &= k_B \left( 1 - \frac{x^2}{12} + O(x^4) \right).\end{aligned}$$

They follow from  $x/(\exp(x) - 1) = 1 - x/2 + x^2/12 + O(x^4)$  and  $\ln(1 - \exp(-x)) = \ln x - x/2 + x^2/24 + O(x^4)$ . The energy and heat-capacity summands stay finite, while the free-energy and entropy summands diverge as  $\ln x = \ln(\varepsilon/k_B T)$ . Near an isolated zero  $\varepsilon_{nk} \propto |\mathbf{k} - \mathbf{k}_0|^a$  with  $a > 0$ , the divergent part of each summand is proportional to  $a \ln |\mathbf{k} - \mathbf{k}_0|$ , and in two dimensions  $\int_{|\mathbf{q}| < q_0} d^2 \mathbf{q} \ln |\mathbf{q}| = \pi q_0^2 (\ln q_0 - \frac{1}{2})$  is finite. The momentum integrals of  $F$  and  $\mathcal{S}$  therefore remain finite in the thermodynamic limit, as stated in *Thermodynamics*. Finiteness of these integrals does not imply that the magnon-gas description is controlled; that condition involves the boson number, treated below.

## Correlation Matrix of the Holstein–Primakoff Bosons

The equal-time correlation matrix  $\langle \hat{\Psi}_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}}^\dagger \rangle$  contains every average of two Holstein–Primakoff operators at momentum  $\mathbf{k}$ . Its blocks are

$$\langle \hat{\Psi}_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}}^\dagger \rangle = \begin{pmatrix} \langle \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger \rangle & \langle \hat{a}_{\mathbf{k}} \hat{a}_{-\mathbf{k}} \rangle \\ \langle \hat{a}_{-\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}^\dagger \rangle & \langle \hat{a}_{-\mathbf{k}}^\dagger \hat{a}_{-\mathbf{k}} \rangle \end{pmatrix},$$

where each block is the  $N_{\text{sub}} \times N_{\text{sub}}$  matrix of averages over the sublattice indices. The average is linear, and the Bogoliubov transformation  $\hat{\Psi}_{\mathbf{k}} = \mathbf{T}_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}}$  has c-number coefficients, so

$$\langle \hat{\Psi}_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}}^\dagger \rangle = \mathbf{T}_{\mathbf{k}} \langle \hat{\Phi}_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}}^\dagger \rangle \mathbf{T}_{\mathbf{k}}^\dagger.$$

The Gibbs state is diagonal in the magnon occupation numbers, so every average of two magnon operators that changes some occupation vanishes. This removes  $\langle \hat{b} \hat{b} \rangle$ ,  $\langle \hat{b}^\dagger \hat{b}^\dagger \rangle$ , and the products of different modes. The remaining averages are  $\langle \hat{b}_{\mathbf{k}n} \hat{b}_{\mathbf{k}n}^\dagger \rangle = 1 + n_{n\mathbf{k}}$  and  $\langle \hat{b}_{-\mathbf{k}n}^\dagger \hat{b}_{-\mathbf{k}n} \rangle = n_{n,-\mathbf{k}}$ , where  $n_{n\mathbf{k}} = n_B(\varepsilon_{n\mathbf{k}})$ . Hence

$$\langle \hat{\Phi}_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}}^\dagger \rangle = \text{diag}(1 + n_{1\mathbf{k}}, \dots, 1 + n_{N_{\text{sub}}\mathbf{k}}, n_{1,-\mathbf{k}}, \dots, n_{N_{\text{sub}},-\mathbf{k}}),$$

which is the matrix  $N_{\mathbf{k}}(0)$  of *Spin Correlations*; each hole entry carries the occupation of its own band at  $-\mathbf{k}$ . In the magnon vacuum all occupations vanish and  $N_{\mathbf{k}}(0) = \text{diag}(I_{N_{\text{sub}}}, 0)$ .

## Boson Number and Sublattice Moment

The boson number of sublattice  $\mu$  averaged over the  $N_{\text{uc}}$  magnetic unit cells is  $\langle \hat{n}_\mu \rangle = N_{\text{uc}}^{-1} \sum_i \langle \hat{a}_{i\mu}^\dagger \hat{a}_{i\mu} \rangle$ . The Fourier phases  $\exp(\pm i\mathbf{k} \cdot \mathbf{r}_{i\mu})$  of the full-position convention cancel in the product of an operator and its adjoint on the same sublattice, and the sum over cells gives  $\sum_i \hat{a}_{i\mu}^\dagger \hat{a}_{i\mu} = \sum_{\mathbf{k}} \hat{a}_{\mathbf{k}\mu}^\dagger \hat{a}_{\mathbf{k}\mu}$ . The diagonal entry  $N_{\text{sub}} + \mu$  of the correlation matrix at  $\mathbf{k}$  is  $\langle \hat{a}_{-\mathbf{k}\mu}^\dagger \hat{a}_{-\mathbf{k}\mu} \rangle$ . The lower block row of  $\mathbf{T}_{\mathbf{k}}$  is  $(\mathbf{Q}_{\mathbf{k}}^*, \mathbf{P}_{-\mathbf{k}}^*)$ , so

$$[\mathbf{T}_{\mathbf{k}} \mathbf{N}_{\mathbf{k}}(0) \mathbf{T}_{\mathbf{k}}^\dagger]_{N_{\text{sub}}+\mu, N_{\text{sub}}+\mu} = \sum_{n=1}^{N_{\text{sub}}} [|\langle \mathbf{Q}_{\mathbf{k}} \rangle_{\mu n}|^2 (1 + n_{n\mathbf{k}}) + |\langle \mathbf{P}_{-\mathbf{k}} \rangle_{\mu n}|^2 n_{n,-\mathbf{k}}].$$

Summing over a momentum set that is closed under inversion and relabeling  $-\mathbf{k} \rightarrow \mathbf{k}$  in the  $\mathbf{P}$  term gives the boson number of *Magnon Observables*,

$$\langle \hat{n}_\mu \rangle = \frac{1}{N_{\text{uc}}} \sum_{\mathbf{k} \in \text{MBZ}} \sum_{n=1}^{N_{\text{sub}}} [|\langle \mathbf{P}_{\mathbf{k}} \rangle_{\mu n}|^2 n_{n\mathbf{k}} + |\langle \mathbf{Q}_{\mathbf{k}} \rangle_{\mu n}|^2 (1 + n_{n\mathbf{k}})].$$

The same result follows directly from  $\hat{a}_{\mathbf{k}\mu} = \sum_n [(\mathbf{P}_{\mathbf{k}})_{\mu n} \hat{b}_{\mathbf{k}n} + (\mathbf{Q}_{-\mathbf{k}})_{\mu n} \hat{b}_{-\mathbf{k}n}^\dagger]$  and the magnon averages above. The longitudinal spin component in the local frame is  $\hat{S}_{i\mu}^0 = S_\mu - \hat{a}_{i\mu}^\dagger \hat{a}_{i\mu}$  to the order of linear spin-wave theory, so the cell-averaged longitudinal moment of sublattice  $\mu$  is  $S_\mu - \langle \hat{n}_\mu \rangle$ .

## References

### Internal Documents

- *Thermodynamics*: states the thermodynamic quantities and the zero-mode limits derived here.
- *Magnon Observables*: states the correlation matrix, the boson number, and the reduced moment derived here.
- *Spin Correlations*: defines the magnon correlation matrix  $\mathbf{N}_{\mathbf{q}}(t)$ , whose equal-time value is derived here.
- *Paraunitary Diagonalization*: supplies the diagonal magnon Hamiltonian and the blocks of  $\mathbf{T}_{\mathbf{k}}$ .
- *Momentum-Space BdG Hamiltonian*: defines the full-position Fourier convention and the Nambu spinor.
- *Notation and Conventions*: fixes the thermodynamic and system-size notation.

### External Sources

- None.

# Topological Magnon Quantities

Source file: docs/lswt/02-observables/topological-magnon-quantities.md — status: draft — updated: 2026-10-01

Two different objects carry topological labels in a spin-wave calculation. A lattice skyrmion number counts how often a classical spin texture wraps the unit sphere, and it depends only on the reference configuration. A magnon Chern number and the magnon thermal Hall conductivity depend on the eigenvectors of the bosonic Bogoliubov–de Gennes (BdG) matrix  $H_k$ , so they inherit its paraunitary normalization, its particle–hole structure, and the Bloch-basis convention used to define  $H_k$ . The band quantities below assume a stable reference configuration, for which  $H_k$  is positive definite at the momenta considered.

## Lattice Skyrmion Number of a Spin Texture

The skyrmion number of a lattice texture is defined on a triangulation of the lattice into oriented elementary triangles. For a triangle with vertices  $I, J, K$  taken counterclockwise and unit spin directions  $\mathbf{m}_I = \mathbf{S}_I/S_I$ , the signed solid angle  $\chi_{IJK} \in (-2\pi, 2\pi)$  spanned by the three directions is fixed by

$$\exp\left(\frac{i\chi_{IJK}}{2}\right) = \frac{1 + \mathbf{m}_I \cdot \mathbf{m}_J + \mathbf{m}_J \cdot \mathbf{m}_K + \mathbf{m}_K \cdot \mathbf{m}_I + i \mathbf{m}_I \cdot (\mathbf{m}_J \times \mathbf{m}_K)}{|1 + \mathbf{m}_I \cdot \mathbf{m}_J + \mathbf{m}_J \cdot \mathbf{m}_K + \mathbf{m}_K \cdot \mathbf{m}_I + i \mathbf{m}_I \cdot (\mathbf{m}_J \times \mathbf{m}_K)|}.$$

The sign of the triple product fixes the orientation of the solid angle, and the complex argument selects the branch, so that  $\chi_{IJK}$  is not restricted to  $(-\pi, \pi)$ . The skyrmion number sums the solid angles over all triangles of one magnetic unit cell,

$$Q_{\text{sk}} = \frac{1}{4\pi} \sum_{\Delta} \chi_{\Delta}. \quad (25)$$

For a texture with the periodicity of the magnetic unit cell, the triangles tile a closed torus and  $Q_{\text{sk}}$  is an integer, except for exceptional configurations in which, on some triangle, the triple product vanishes while the real part  $1 + \mathbf{m}_I \cdot \mathbf{m}_J + \mathbf{m}_J \cdot \mathbf{m}_K + \mathbf{m}_K \cdot \mathbf{m}_I$  is not positive. Such a triangle has three directions on one great circle that are not contained in a half circle, or two antiparallel directions; its solid angle is  $\pm 2\pi$  or undefined, and  $Q_{\text{sk}}$  is not defined for that configuration. On the triangular lattice the elementary triangles are the up- and down-pointing plaquettes, two per crystallographic unit cell. The skyrmion number is a property of the classical reference configuration and does not require the spin-wave expansion.

## Berry Curvature of Bosonic Bogoliubov Bands

The paraunitary transformation  $T_k$  diagonalizes the  $2N_{\text{sub}} \times 2N_{\text{sub}}$  matrix  $H_k$  while preserving the Nambu metric,

$$T_k^\dagger \Sigma_3 T_k = \Sigma_3, \quad T_k^\dagger H_k T_k = \text{diag}(\varepsilon_{1k}, \dots, \varepsilon_{N_{\text{sub}}k}, \varepsilon_{1,-k}, \dots, \varepsilon_{N_{\text{sub}},-k}).$$

The first  $N_{\text{sub}}$  columns of  $T_k$  belong to the physical magnon bands  $n = 1, \dots, N_{\text{sub}}$ , and the remaining columns are their particle-hole partners. We use the signed energies  $\lambda_{pk} = (\Sigma_3)_{pp} (T_k^\dagger H_k T_k)_{pp}$  for  $p = 1, \dots, 2N_{\text{sub}}$ , which equal  $\varepsilon_{nk}$  on the particle columns and  $-\varepsilon_{n,-k}$  on the hole columns. With the paraunitary Berry connection  $A_{nk} = i(\Sigma_3)_{nn} [T_k^\dagger \Sigma_3 \nabla_k T_k]_{nn}$ , the Berry curvature  $\Omega_{nk} = \partial_{k_x} A_{nk}^y - \partial_{k_y} A_{nk}^x$  of a physical band takes the Kubo form

$$\Omega_{nk} = -2 \text{Im} \sum_{\substack{p=1 \\ p \neq n}}^{2N_{\text{sub}}} (\Sigma_3)_{nn} (\Sigma_3)_{pp} \frac{[T_k^\dagger (\partial_{k_x} H_k) T_k]_{np} [T_k^\dagger (\partial_{k_y} H_k) T_k]_{pn}}{(\lambda_{nk} - \lambda_{pk})^2}. \quad (26)$$

The sum runs over all  $2N_{\text{sub}}$  columns. Terms with another particle band  $p = m \leq N_{\text{sub}}$  have the denominator  $(\varepsilon_{nk} - \varepsilon_{mk})^2$ , whereas terms with a hole column have  $(\varepsilon_{nk} + \varepsilon_{m,-k})^2$  and carry the anomalous (pairing) couplings. When  $H_k$  conserves the boson number, the hole columns decouple and the expression reduces to the Berry curvature of a Hermitian band problem. The curvature of band  $n$  is defined where the band is separated from every other signed energy. It is undefined, and generically divergent, where two physical bands touch and where a particle energy meets a hole energy,  $\varepsilon_{nk} + \varepsilon_{m,-k} \rightarrow 0$ ; for a reciprocal spectrum the latter reduces to  $\varepsilon_{nk} \rightarrow 0$ .

The pointwise curvature depends on the Bloch-basis convention used for  $H_k$ . The full-position convention attaches the phase  $\exp(i\mathbf{k} \cdot (\mathbf{R}_i + \boldsymbol{\tau}_a))$  to the boson on sublattice  $a$  of cell  $i$ , whereas the cell convention attaches  $\exp(i\mathbf{k} \cdot \mathbf{R}_i)$ . The two are related by the sublattice-dependent phase matrix  $\text{diag} \exp(\pm i\mathbf{k} \cdot \boldsymbol{\tau}_a)$ , applied equally to the particle and hole components. This relation is not a gauge transformation. A gauge transformation multiplies each band by its own phase,  $t_{nk} \rightarrow \exp(i\chi_n(\mathbf{k})) t_{nk}$ , and leaves every observable unchanged; the change of convention instead replaces the position operator  $\sum_i \mathbf{r}_i \hat{n}_i$  by  $\sum_i \mathbf{R}_i \hat{n}_i$ , which describes a system in which every site sits at its cell origin. It adds to  $A_{nk}$  the term  $\pm \sum_a w_{na}(\mathbf{k}) \boldsymbol{\tau}_a$ , where  $w_{na}$  is the  $\Sigma_3$ -weighted weight of band  $n$  on sublattice  $a$ , and changes  $\Omega_{nk}$  by the curl of that term. Because  $w_{na}(\mathbf{k})$  is periodic over the magnetic Brillouin zone (MBZ), the curl integrates to zero, so the Chern number is the same in both conventions, while plots of  $\Omega_{nk}$  differ. A momentum integral that weights  $\Omega_{nk}$  by a nonconstant function of  $\varepsilon_{nk}$  is not protected by this argument. For the thermal Hall conductivity below, integration by parts leaves a difference proportional to  $\sum_a \hat{\mathbf{z}} \cdot (\mathbf{J}_a \times \boldsymbol{\tau}_a)$ , where  $\mathbf{J}_a$  is the momentum average of  $\sum_n c_2' w_{na} \nabla_k \varepsilon_{nk}$ , the weighted group velocity carried by sublattice  $a$ ; it vanishes in general only when a symmetry maps each sublattice onto itself, for example a threefold or twofold rotation about a site. The thermal Hall response is defined through the position operator of the actual site positions, which enters the energy magnetization, and the full-position convention is the one whose Bloch states correspond to that operator. The cell-convention value is not a property of the system: relabeling the sites of one sublattice into a neighboring cell,  $\boldsymbol{\tau}_a \rightarrow \boldsymbol{\tau}_a + \mathbf{R}$ , describes the same system but changes that value by a term proportional to  $\hat{\mathbf{z}} \cdot (\mathbf{J}_a \times \mathbf{R})$ , whereas the full-position value is unchanged. We therefore evaluate all band quantities in the full-position convention. The remaining approximation is the tight-binding assumption that the magnon position operator is diagonal at the site positions, which is a property of the model rather than a choice of convention.

## Chern Number of an Isolated Magnon Band

For a physical band  $n$  separated from all other signed energies throughout the MBZ, the Chern number is the integral of the Berry curvature over the MBZ,

$$C_n = \frac{1}{2\pi} \int_{\text{MBZ}} \Omega_{nk} d^2k \approx \frac{1}{2\pi} \frac{A_{\text{MBZ}}}{N_k} \sum_k \Omega_{nk}, \quad (27)$$

where the discrete form uses a uniform mesh of  $N_k$  momenta and  $A_{\text{MBZ}} = (2\pi)^2/A_{\text{uc}}$  is the area of the MBZ for a magnetic unit cell of area  $A_{\text{uc}}$ . The integral is an integer for an isolated band, whereas the discrete sum approaches it as the mesh is refined and converges slowly where the curvature is concentrated near a small gap. The lattice link-variable construction of Fukui, Hatsugai, and Suzuki evaluates the same invariant from the eigenvectors on neighboring mesh points; for bosonic bands the link variables use the paraunitary product  $t_{nk}^\dagger \Sigma_3 t_{nk'}$  of the columns  $t_{nk}$  of  $T_k$ . That construction returns an integer on every mesh, including meshes that do not resolve a gap closing, so an integer value by itself does not establish that the band is isolated. Neither construction defines a Chern number for bands that touch; a group of touching bands carries only a combined invariant, which is not treated here.

## Magnon Thermal Hall Conductivity

The transverse thermal conductivity of a magnetic layer follows from the Berry curvature of the thermally occupied magnon bands. For isolated bands, the conductivity per layer is

$$\kappa_{xy}^{2D} = -\frac{k_B^2 T}{\hbar} \frac{1}{N_k A_{\text{uc}}} \sum_k \sum_{n=1}^{N_{\text{sub}}} c_2(n_B(\varepsilon_{nk})) \Omega_{nk}, \quad (28)$$

with the Bose–Einstein distribution  $n_B(\varepsilon) = [\exp(\varepsilon/k_B T) - 1]^{-1}$  and the weight function

$$c_2(x) = \int_0^x \left( \ln \frac{1+t}{t} \right)^2 dt = (1+x) \left( \ln \frac{1+x}{x} \right)^2 - (\ln x)^2 - 2 \text{Li}_2(-x), \quad (29)$$

where  $\text{Li}_2(z) = -\int_0^z \ln(1-t) t^{-1} dt$  is the dilogarithm. The weight vanishes as  $x \rightarrow 0$  and approaches  $\pi^2/3$  as  $x \rightarrow \infty$ , so thermally unoccupied bands do not contribute. The factor  $1/(N_k A_{\text{uc}})$  turns the momentum sum into  $\int_{\text{MBZ}} d^2k/(2\pi)^2$ ; the Berry curvature and the cell area carry the same squared length unit, which cancels, and  $\kappa_{xy}^{2D}$  has the unit of  $k_B^2 T/\hbar$ . For a stack of equivalent, independent layers with interlayer spacing  $d$ , the three-dimensional conductivity is  $\kappa_{xy} = \kappa_{xy}^{2D}/d$ .

The band sum in the thermal Hall conductivity requires the individual curvatures only through the combination  $\sum_n c_2(n_B(\varepsilon_{nk})) \Omega_{nk}$  at each momentum. Inserting the Kubo form and pairing the terms of two particle bands  $n$  and  $m$  gives

$$\sum_{n=1}^{N_{\text{sub}}} c_2^{(n)} \Omega_{nk} = -2 \sum_{n < m} (c_2^{(n)} - c_2^{(m)}) \frac{\text{Im}([T_k^\dagger \partial_{k_x} H_k T_k]_{nm} [T_k^\dagger \partial_{k_y} H_k T_k]_{mn})}{(\varepsilon_{nk} - \varepsilon_{mk})^2} + \sum_{n=1}^{N_{\text{sub}}} c_2^{(n)} \Omega_{nk}^{\text{ph}}, \quad (30)$$

where  $c_2^{(n)} = c_2(n_B(\varepsilon_{nk}))$  and  $\Omega_{nk}^{\text{ph}}$  collects the hole-column terms of the Kubo form. The weight difference vanishes for exactly degenerate particle bands and is proportional to the energy difference near a crossing, so this combination remains finite where individual curvatures are undefined. Only a degeneracy between a particle band and a hole column, which requires  $\varepsilon_{nk} \rightarrow 0$ , leaves it undefined.

The primary note writes the weight as  $c_2(n_B) - \pi^2/3$  and normalizes by a system volume  $V$  without  $\hbar$ . The volume factor is replaced here by the per-layer area  $N_k A_{\text{uc}}$  and the explicit  $\hbar$ . The two weights differ by a constant. With the discrete MBZ sum written as an integral, the conductivity obtained with the primary-note weight exceeds the value from the per-layer formula by

$$\frac{\pi^2}{3} \frac{k_B^2 T}{\hbar} \int_{\text{MBZ}} \frac{d^2 \mathbf{k}}{(2\pi)^2} \sum_{n=1}^{N_{\text{sub}}} \Omega_{nk} = \frac{\pi k_B^2 T}{6\hbar} \sum_{n=1}^{N_{\text{sub}}} C_n,$$

so the two forms coincide when the Chern numbers of the physical bands sum to zero. For a Hamiltonian matrix  $H_k$  that is positive definite on the whole MBZ, the sum does vanish. The family  $H_{k,\lambda} = (1-\lambda)H_k + \lambda\mathbb{1}$ ,  $0 \leq \lambda \leq 1$ , stays positive definite, so its particle and hole bands never touch. The total Chern number of the particle bands can change only at such a touching, so it equals its value at  $\lambda = 1$ , where  $T_k = \mathbb{1}$  and every curvature vanishes (Shindou et al., Eq. (29)). The two weights therefore give the same conductivity. This note uses the weight without the constant, as Matsumoto and Murakami and Neumann et al. do; it gives thermally unoccupied bands zero weight directly. At high temperature every weight approaches  $\pi^2/3$ , so this form gives  $\kappa_{xy}/T \rightarrow -(\pi k_B^2/6\hbar) \sum_n C_n$ , which vanishes by the sum rule above; the primary-note weight approaches zero in that limit directly. When  $H_k$  is only positive semidefinite, as with Goldstone modes, particle and hole columns meet at zero energy and the argument does not apply as stated; that case is left open.

## References

### Internal Documents

- *Paraunitary Diagonalization*: defines  $T_k$ , the Nambu metric, and the stability condition assumed for the band quantities.
- *Momentum-Space BdG Hamiltonian*: defines  $H_k$  and its full-position Fourier convention, which fixes the Bloch basis of the pointwise Berry curvature.
- *Magnon Observables*: supplies the magnon bands and occupations used by the thermal Hall conductivity.
- *Classical Order and Local Frame*: defines the classical spin directions used by the skyrmion number.
- *Notation and Conventions*: fixes  $\Sigma_3$ ,  $\varepsilon_{nk}$ ,  $N_{\text{sub}}$ , and the MBZ notation, and lists the thermal-Hall units as not yet fixed.

### External Sources

- B. Berg and M. Lüscher, “Definition and Statistical Distributions of a Topological Number in the Lattice O(3) Sigma-Model,” *Nuclear Physics B* **190**, 412-424 (1981), [doi:10.1016/0550-3213\(81\)90568-X](https://doi.org/10.1016/0550-3213(81)90568-X): signed solid angle of a lattice triangle and the integer lattice topological number.
- R. Shindou, R. Matsumoto, S. Murakami, and J. Ohe, “Topological chiral magnonic edge mode in a magnonic crystal,” *Physical Review B* **87**, 174427 (2013), [doi:10.1103/PhysRevB.87.174427](https://doi.org/10.1103/PhysRevB.87.174427):

Berry curvature and Chern number of bosonic BdG bands with the paraunitary normalization; Eq. (29) proves that the Chern numbers of the particle bands sum to zero for a positive definite Hamiltonian matrix.

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- T. Fukui, Y. Hatsugai, and H. Suzuki, “Chern Numbers in Discretized Brillouin Zone: Efficient Method of Computing (Spin) Hall Conductances,” *Journal of the Physical Society of Japan* **74**, 1674-1677 (2005), [doi:10.1143/JPSJ.74.1674](https://doi.org/10.1143/JPSJ.74.1674): lattice link-variable construction of Chern numbers.
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# Spin Correlations

Source file: docs/lswt/02-observables/spin-correlations.md — status: draft — updated: 2026-10-01

Neutron scattering and other probes of magnetic excitations measure correlations between spin components at different sites and times. This document defines the momentum- and time-resolved spin correlation function and evaluates it in LSWT. The spin operator is expanded in the local frame of the classical reference configuration, the transverse part is expressed through the magnon spinor and the paraunitary transformation, and the longitudinal part supplies an elastic contribution at magnetic reciprocal-lattice vectors. The static and dynamic structure factors and the spectral function built from this correlation function are treated in *Structure Factor and Spectral Function*.

## Correlation Function and Its Symmetries

We set  $\hbar = 1$  in the time evolution, so that frequencies and magnon energies share energy units. Spin operators evolve in the Heisenberg picture,  $\hat{S}_I^\alpha(t) = \exp(i\hat{H}t)\hat{S}_I^\alpha \exp(-i\hat{H}t)$ , and  $\langle \dots \rangle$  is the thermal average in the Gibbs state. The external momentum  $q$  is not restricted to the magnetic Brillouin zone (MBZ). The spin Fourier components use the full site positions  $r_I$  and are normalized per site,

$$\hat{S}_q^\alpha = \frac{1}{\sqrt{N_{\text{site}}}} \sum_I \exp(-iq \cdot r_I) \hat{S}_I^\alpha, \quad (31)$$

where  $\alpha$  labels a laboratory Cartesian component. The real-time spin correlation function is the  $3 \times 3$  matrix

$$C^{\alpha\beta}(q, t) = \langle \hat{S}_q^\alpha(t) \hat{S}_{-q}^\beta(0) \rangle = \frac{1}{N_{\text{site}}} \sum_{I,J} \exp(-iq \cdot (r_I - r_J)) \langle \hat{S}_I^\alpha(t) \hat{S}_J^\beta(0) \rangle. \quad (32)$$

The correlation function depends on the positions only through the differences  $r_I - r_J$ . It is therefore unchanged under  $q \rightarrow q + G$  for every vector with  $G \cdot (r_I - r_J) \in 2\pi\mathbb{Z}$  for all pairs of sites, that is, for the reciprocal-lattice vectors of the lattice generated by the site differences. A magnetic reciprocal-lattice vector alone does not satisfy this condition when the magnetic unit cell contains sites that are not related by magnetic lattice translations.

Two relations follow from the definition. Because the Cartesian components are Hermitian,  $(\hat{S}_q^\alpha)^\dagger = \hat{S}_{-q}^\alpha$ . Because the Gibbs state commutes with the time evolution,  $\langle \hat{A}(0)\hat{B}(t) \rangle = \langle \hat{A}(-t)\hat{B}(0) \rangle$ . Taking the complex conjugate of the average,  $[\langle \hat{S}_q^\alpha(t) \hat{S}_{-q}^\beta \rangle]^* = \langle \hat{S}_q^\beta \hat{S}_{-q}^\alpha(t) \rangle = \langle \hat{S}_q^\beta(-t) \hat{S}_{-q}^\alpha \rangle$ , so together they give

$$[C^{\alpha\beta}(\mathbf{q}, t)]^* = C^{\beta\alpha}(\mathbf{q}, -t). \quad (33)$$

For a complex combination  $\hat{S}^c = \sum_{\alpha} c_{\alpha} \hat{S}^{\alpha}$ , such as a laboratory ladder component, the Hermitian conjugate is  $(\hat{S}_q^c)^{\dagger} = \hat{S}_{-q}^{c*}$ . The correlation of such components follows from the Cartesian matrix by linearity,  $\langle \hat{S}_q^c(t) \hat{S}_{-q}^d(0) \rangle = \sum_{\alpha\beta} c_{\alpha} d_{\beta} C^{\alpha\beta}(\mathbf{q}, t)$ , and the conjugation relation then exchanges  $c$  and  $d$  with their complex conjugates.

## Linear Spin-Wave Expansion of the Spin Operator

The local frame of site  $I$  has the circular vectors  $\mathbf{e}_I^{\pm} = R_I(\mathbf{e}_x \pm i\mathbf{e}_y)/\sqrt{2}$  and the longitudinal vector  $\mathbf{e}_I^0 = \mathbf{n}_I$ , defined in *Classical Order and Local Frame*. With the local ladder operators  $\hat{S}_I^{\pm} = \hat{S}_I^x \pm i\hat{S}_I^y$ , the spin operator reads  $\hat{S}_I = (\hat{S}_I^+ \mathbf{e}_I^- + \hat{S}_I^- \mathbf{e}_I^+)/\sqrt{2} + \hat{S}_I^0 \mathbf{e}_I^0$ . Its laboratory component  $\alpha$  is therefore

$$\hat{S}_I^{\alpha} = \frac{1}{\sqrt{2}} \left( \bar{u}_I^{\alpha} \hat{S}_I^+ + u_I^{\alpha} \hat{S}_I^- \right) + v_I^{\alpha} \hat{S}_I^0, \quad u_I^{\alpha} = \mathbf{e}_{\alpha} \cdot \mathbf{e}_I^+, \quad \bar{u}_I^{\alpha} = (u_I^{\alpha})^* = \mathbf{e}_{\alpha} \cdot \mathbf{e}_I^-, \quad v_I^{\alpha} = \mathbf{e}_{\alpha} \cdot \mathbf{n}_I.$$

The coefficients  $u_I^{\alpha}$  and  $v_I^{\alpha}$  are the laboratory components of the local basis vectors. Inserting the leading Holstein–Primakoff forms  $\hat{S}_I^+ \simeq \sqrt{2S_I} \hat{a}_I$ ,  $\hat{S}_I^- \simeq \sqrt{2S_I} \hat{a}_I^{\dagger}$ , and  $\hat{S}_I^0 = S_I - \hat{n}_I$  of *Holstein–Primakoff Expansion* gives

$$\hat{S}_I^{\alpha} \simeq v_I^{\alpha} (S_I - \hat{n}_I) + \sqrt{S_I} (\bar{u}_I^{\alpha} \hat{a}_I + u_I^{\alpha} \hat{a}_I^{\dagger}). \quad (34)$$

The longitudinal part contains the classical moment, of order  $S$ , and the boson number. The transverse part is linear in the bosons and of order  $S^{1/2}$ . The correlation function is kept to order  $S$ . At this order it contains two pieces: the product of the ordered moments, of order  $S^2$ , together with its reduction by  $\langle \hat{n}_I \rangle$ , of order  $S$ ; and the transverse correlation, of order  $S$ . Cross terms between the transverse part and the constant  $S_I$  vanish because  $\langle \hat{a}_I \rangle = 0$  in the Gibbs state of a quadratic Hamiltonian without linear terms. Cross terms between the transverse part and  $\hat{n}_I$  contain an odd number of boson operators and vanish by Wick’s theorem. The connected longitudinal correlation  $\langle \hat{n}_I(t) \hat{n}_J(0) \rangle - \langle \hat{n}_I \rangle \langle \hat{n}_J \rangle$  is of order  $S^0$ ; it describes two-magnon scattering and is omitted, as are the  $O(S^0)$  corrections to the transverse part from the next Holstein–Primakoff terms.

## Transverse Correlations in Momentum Space

We write sites as  $I = (i, \mu)$  with  $\mathbf{r}_{i\mu} = \mathbf{R}_i + \delta_{\mu}$ , and use the full-position Fourier convention of *Momentum-Space BdG Hamiltonian*. The inverse transform  $\hat{a}_{\mathbf{q}\mu} = N_{\text{uc}}^{-1/2} \sum_i \exp(-i\mathbf{q} \cdot \mathbf{r}_{i\mu}) \hat{a}_{i\mu}$  defines the boson at any momentum. Momenta that differ by a magnetic reciprocal-lattice vector  $\mathbf{G}$  are related by  $\hat{a}_{\mathbf{q}+\mathbf{G},\mu} = \exp(-i\mathbf{G} \cdot \delta_{\mu}) \hat{a}_{\mathbf{q}\mu}$ . The Nambu spinor  $\hat{\Psi}_{\mathbf{q}}$ , the BdG matrix  $\mathbf{H}_{\mathbf{q}}$ , and the paraunitary matrix  $\mathbf{T}_{\mathbf{q}}$  are evaluated at the unfolded momentum  $\mathbf{q}$ . The energies  $\varepsilon_{n\mathbf{q}}$  are periodic over the MBZ, while  $\hat{\Psi}_{\mathbf{q}+\mathbf{G}} = \Lambda_{\mathbf{G}} \hat{\Psi}_{\mathbf{q}}$  and  $\mathbf{T}_{\mathbf{q}+\mathbf{G}} = \Lambda_{\mathbf{G}} \mathbf{T}_{\mathbf{q}}$  with  $\Lambda_{\mathbf{G}} = \mathbf{I}_2 \otimes \text{diag}(\exp(-i\mathbf{G} \cdot \delta_{\mu}))$ , the same phase in both Nambu blocks.

For a periodic reference configuration the coefficients depend only on the sublattice,  $u_I^{\alpha} = u_{\mu}^{\alpha}$  and  $v_I^{\alpha} = v_{\mu}^{\alpha}$ . Summing Eq. (34) with the Fourier phases gives the transverse part of the spin component,

$$\hat{S}_{q,\perp}^\alpha = V^\alpha \hat{\Psi}_q, \quad V^\alpha = \frac{1}{\sqrt{N_{\text{sub}}}} \left( \sqrt{S_1} \tilde{u}_1^\alpha, \dots, \sqrt{S_{N_{\text{sub}}}} \tilde{u}_{N_{\text{sub}}}^\alpha, \sqrt{S_1} u_1^\alpha, \dots, \sqrt{S_{N_{\text{sub}}}} u_{N_{\text{sub}}}^\alpha \right), \quad (35)$$

where the spin vertex  $V^\alpha$  is a row vector with  $2N_{\text{sub}}$  entries. The vertex does not depend on the momentum; all momentum dependence, including the sublattice phases, resides in  $\hat{\Psi}_q$ . Because  $\hat{S}_{-q}^\beta = (\hat{S}_q^\beta)^\dagger$ , the second operator of the correlation function is  $\hat{S}_{-q,\perp}^\beta = \hat{\Psi}_q^\dagger (V^\beta)^\dagger$ .

The magnon spinor  $\hat{\Phi}_q = T_q^{-1} \hat{\Psi}_q$  evolves as  $\hat{b}_{qn}(t) = \exp(-i\varepsilon_{qn}t) \hat{b}_{qn}$  and  $\hat{b}_{-qn}^\dagger(t) = \exp(i\varepsilon_{n,-q}t) \hat{b}_{-qn}^\dagger$ . In the Gibbs state its two-time correlation matrix is diagonal:

$$N_q(t) = \langle \hat{\Phi}_q(t) \hat{\Phi}_q^\dagger(0) \rangle = \text{diag} \left( (1 + n_{1q}) \exp(-i\varepsilon_{1q}t), \dots, n_{1,-q} \exp(i\varepsilon_{1,-q}t), \dots \right), \quad (36)$$

with  $n_{nq} = n_B(\varepsilon_{nq})$  from *Thermodynamics*. The particle entries carry  $1 + n$  because  $\langle \hat{b} \hat{b}^\dagger \rangle = 1 + \langle \hat{b}^\dagger \hat{b} \rangle$ ; the hole entries carry  $n$ . The anomalous averages  $\langle \hat{b} \hat{b} \rangle$  vanish. The transverse correlation function is then

$$C_{\perp}^{\alpha\beta}(q, t) = V^\alpha T_q N_q(t) T_q^\dagger (V^\beta)^\dagger = \sum_{p=1}^{2N_{\text{sub}}} (V^\alpha T_q)_p (V^\beta T_q)_p^* [N_q(t)]_{pp}. \quad (37)$$

Each column  $p$  of  $T_q$  contributes one term with the amplitude  $(V^\alpha T_q)_p$ . A particle column describes the creation of a magnon  $(q, n)$  by  $\hat{S}_{-q}^\beta$ , and a hole column the absorption of a thermally excited magnon  $(-q, n)$ .

The sublattice-resolved correlation  $C_{\mu\nu}^{\alpha\beta}$ , whose sum over  $\mu$  and  $\nu$  gives  $C^{\alpha\beta}$ , collects the terms of Eq. (37) in which the entries of  $V^\alpha$  and  $V^\beta$  belong to sublattices  $\mu$  and  $\nu$ . It is the sum of the four entries  $(\mu, \nu)$ ,  $(\mu, N_{\text{sub}} + \nu)$ ,  $(N_{\text{sub}} + \mu, \nu)$ , and  $(N_{\text{sub}} + \mu, N_{\text{sub}} + \nu)$  of the matrix  $\text{diag}(V^\alpha) T_q N_q(t) T_q^\dagger \text{diag}(V^\beta)^\dagger$ , where  $\text{diag}(V^\alpha)$  places the entries of  $V^\alpha$  on the diagonal. In the full-position convention no additional sublattice phase  $\exp(-iq \cdot (\delta_\mu - \delta_\nu))$  appears, since those phases are already contained in  $\hat{\Psi}_q$ .

## Elastic Correlations at Magnetic Bragg Vectors

The longitudinal part of Eq. (34) has the thermal average  $\langle \hat{S}_I^\alpha \rangle = v_\mu^\alpha (S_\mu - \langle \hat{n}_\mu \rangle) = m_\mu^\alpha$ , the ordered moment of sublattice  $\mu$  reduced by the boson number  $\langle \hat{n}_\mu \rangle$  of *Magnon Observables*. Its Fourier sum over the  $N_{\text{uc}}$  cells is  $N_{\text{uc}} \delta_{q,G}$ , a Kronecker delta that selects the magnetic reciprocal-lattice vectors  $G$ . The product of the averages gives the elastic correlation

$$C_{\text{el}}^{\alpha\beta}(q) = \langle \hat{S}_q^\alpha \rangle \langle \hat{S}_{-q}^\beta \rangle = N_{\text{site}} \sum_G \delta_{q,G} \frac{F^\alpha(G) F^\beta(G)^*}{N_{\text{sub}}^2}, \quad F^\alpha(G) = \sum_{\mu=1}^{N_{\text{sub}}} \exp(-iG \cdot \delta_\mu) m_\mu^\alpha, \quad (38)$$

where  $F^\alpha(G)$  is the magnetic structure factor of the magnetic unit cell. The elastic correlation does not depend on time. It is proportional to  $N_{\text{site}}$ , reflecting long-range order, while the transverse correlation is of order one per site. To order  $S$ , the product of the reduced moments contains the classical term  $S_\mu S_\nu v_\mu^\alpha v_\nu^\beta$  and the reductions  $-v_\mu^\alpha v_\nu^\beta (S_\mu \langle \hat{n}_\nu \rangle + S_\nu \langle \hat{n}_\mu \rangle)$ ; the remaining product  $\langle \hat{n}_\mu \rangle \langle \hat{n}_\nu \rangle$  is of order  $S^0$  and is retained only because the moments appear in factorized form.

To the order kept, the correlation function is

$$C^{\alpha\beta}(\mathbf{q}, t) = C_{\text{el}}^{\alpha\beta}(\mathbf{q}) + C_{\perp}^{\alpha\beta}(\mathbf{q}, t). \quad (39)$$

## Equal-Time Correlations in Real Space

The equal-time correlation between two sites follows from the same expansion without the momentum transform. For  $I = (i, \mu)$  and  $J = (j, \nu)$ ,

$$\langle \hat{S}_I^{\alpha} \hat{S}_J^{\beta} \rangle \simeq m_{\mu}^{\alpha} m_{\nu}^{\beta} + \sqrt{S_{\mu} S_{\nu}} \frac{1}{N_{\text{uc}}} \sum_{\mathbf{k} \in \text{MBZ}} \exp(i\mathbf{k} \cdot (\mathbf{r}_I - \mathbf{r}_J)) w_{\mu}^{\alpha} [\mathbf{T}_{\mathbf{k}} \mathbf{N}_{\mathbf{k}}(0) \mathbf{T}_{\mathbf{k}}^{\dagger}]_{\{\mu\}, \{\nu\}} (w_{\nu}^{\beta})^{\dagger},$$

where  $w_{\mu}^{\alpha} = (\bar{u}_{\mu}^{\alpha}, u_{\mu}^{\alpha})$  and  $[\dots]_{\{\mu\}, \{\nu\}}$  is the  $2 \times 2$  submatrix with rows  $\mu, N_{\text{sub}} + \mu$  and columns  $\nu, N_{\text{sub}} + \nu$ . The connected longitudinal part  $\langle \hat{n}_I \hat{n}_J \rangle - \langle \hat{n}_I \rangle \langle \hat{n}_J \rangle$ , of order  $S^0$ , is omitted for every pair of sites, including  $I = J$ . Correlations of local-frame components follow by replacing the laboratory vector  $\mathbf{e}_{\alpha}$  with a local basis vector of the respective site in the definitions of  $u_I^{\alpha}$  and  $v_I^{\alpha}$ .

## References

### Internal Documents

- *Classical Order and Local Frame*: defines  $\mathbf{R}_I$ ,  $\mathbf{n}_I$ , and the local circular vectors  $\mathbf{e}_I^{\pm}$ .
- *Holstein–Primakoff Expansion*: supplies the leading boson forms of the local spin operators.
- *Momentum-Space BdG Hamiltonian*: defines the full-position Fourier convention and the Nambu spinor.
- *Paraunitary Diagonalization*: supplies  $\mathbf{T}_{\mathbf{q}}$  and the magnon energies.
- *Thermodynamics*: defines the Bose–Einstein occupations.
- *Magnon Observables*: owns the boson numbers and the reduced moments  $m_{\mu}^{\alpha}$ .
- *Structure Factor and Spectral Function*: builds the scattering observables from  $C^{\alpha\beta}(\mathbf{q}, t)$ .
- *Notation and Conventions*: distinguishes the external momentum  $\mathbf{q}$  from the magnon momentum  $\mathbf{k}$  and reserves the spin-vertex symbol.

### External Sources

- None.

# Structure Factor and Spectral Function

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The spin correlation function  $C^{\alpha\beta}(\mathbf{q}, t)$  of *Spin Correlations* determines the momentum- and energy-resolved observables of magnetic excitations. Its equal-time value is the static structure factor, its Fourier transform in time is the dynamic structure factor measured by inelastic neutron scattering, and the Fourier transform of the corresponding commutator is the spectral function, which determines the retarded response. This document derives these quantities in LSWT, where each magnon mode contributes a delta function in energy with a weight fixed by the paraunitary transformation. We keep  $\hbar = 1$  and the per-site normalization of *Spin Correlations*.

## One-Magnon Weights

The transverse correlation of *Spin Correlations*,  $C_{\perp}^{\alpha\beta}(\mathbf{q}, t) = \mathbf{V}^{\alpha} \mathbf{T}_{\mathbf{q}} \mathbf{N}_{\mathbf{q}}(t) \mathbf{T}_{\mathbf{q}}^{\dagger} (\mathbf{V}^{\beta})^{\dagger}$ , is a sum over the columns of  $\mathbf{T}_{\mathbf{q}}$ . For a particle column  $n = 1, \dots, N_{\text{sub}}$  we define the one-magnon weight

$$W_n^{\alpha\beta}(\mathbf{q}) = (\mathbf{V}^{\alpha} \mathbf{T}_{\mathbf{q}})_n (\mathbf{V}^{\beta} \mathbf{T}_{\mathbf{q}})_n^*, \quad (40)$$

which is a Hermitian, positive-semidefinite  $3 \times 3$  matrix of rank one in  $(\alpha, \beta)$ . The phase and degeneracy freedom of  $\mathbf{T}_{\mathbf{q}}$  described in *Paraunitary Diagonalization* leaves  $W_n^{\alpha\beta}$  unchanged for a nondegenerate band and leaves the sum over a degenerate set unchanged.

The hole columns are determined by the particle columns at  $-\mathbf{q}$ . The hole column  $N_{\text{sub}} + n$  of  $\mathbf{T}_{\mathbf{q}}$  can be chosen as  $\Sigma_1$  times the complex conjugate of the particle column  $n$  of  $\mathbf{T}_{-\mathbf{q}}$ , and the spin vertex  $\mathbf{V}^{\alpha}$  of *Spin Correlations* satisfies  $\mathbf{V}^{\alpha} \Sigma_1 = (\mathbf{V}^{\alpha})^*$ . Hence  $(\mathbf{V}^{\alpha} \mathbf{T}_{\mathbf{q}})_{N_{\text{sub}}+n} = [(\mathbf{V}^{\alpha} \mathbf{T}_{-\mathbf{q}})_n]^*$ , and the hole term of mode  $n$  carries the weight  $W_n^{\beta\alpha}(-\mathbf{q})$  at the energy  $\varepsilon_{n,-\mathbf{q}}$ .

## Static Structure Factor

The static structure factor is the equal-time correlation function,

$$\mathcal{S}^{\alpha\beta}(\mathbf{q}) = C^{\alpha\beta}(\mathbf{q}, 0) = C_{\text{el}}^{\alpha\beta}(\mathbf{q}) + \sum_{n=1}^{N_{\text{sub}}} \left[ (1 + n_{n\mathbf{q}}) W_n^{\alpha\beta}(\mathbf{q}) + n_{n,-\mathbf{q}} W_n^{\beta\alpha}(-\mathbf{q}) \right], \quad (41)$$

with  $n_{n\mathbf{q}} = n_{\text{B}}(\varepsilon_{n\mathbf{q}})$ . The elastic part  $C_{\text{el}}^{\alpha\beta}(\mathbf{q})$ , given in *Spin Correlations*, is present only at magnetic reciprocal-lattice vectors and is proportional to  $N_{\text{site}}$ ; the inelastic part is of order one per site. At zero

temperature only the particle weights remain. Because the correlation function includes all energy transfers,  $\zeta^{\alpha\beta}(\mathbf{q})$  is also the frequency integral of the dynamic structure factor below.

## Dynamic Structure Factor

The dynamic structure factor is the time Fourier transform of the correlation function,

$$\zeta^{\alpha\beta}(\mathbf{q}, \omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt \exp(i\omega t) C^{\alpha\beta}(\mathbf{q}, t). \quad (42)$$

The elastic part is time independent and transforms into  $\delta(\omega)$ . In the transverse part, the particle entries of  $N_{\mathbf{q}}(t)$  oscillate as  $\exp(-i\varepsilon_{n\mathbf{q}}t)$  and the hole entries as  $\exp(+i\varepsilon_{n,-\mathbf{q}}t)$ , so they transform into delta functions at positive and negative energy transfer, respectively:

$$\zeta^{\alpha\beta}(\mathbf{q}, \omega) = C_{\text{el}}^{\alpha\beta}(\mathbf{q}) \delta(\omega) + \sum_{n=1}^{N_{\text{sub}}} \left[ (1 + n_{n\mathbf{q}}) W_n^{\alpha\beta}(\mathbf{q}) \delta(\omega - \varepsilon_{n\mathbf{q}}) + n_{n,-\mathbf{q}} W_n^{\beta\alpha}(-\mathbf{q}) \delta(\omega + \varepsilon_{n,-\mathbf{q}}) \right]. \quad (43)$$

Positive  $\omega$  corresponds to energy transferred to the magnet by creating a magnon  $(\mathbf{q}, n)$ ; negative  $\omega$  corresponds to the absorption of a thermally excited magnon  $(-\mathbf{q}, n)$ . The ratio  $n_{\text{B}}(\varepsilon)/[1 + n_{\text{B}}(\varepsilon)] = \exp(-\beta\varepsilon)$  gives detailed balance,

$$\zeta^{\beta\alpha}(-\mathbf{q}, -\omega) = \exp(-\beta\omega) \zeta^{\alpha\beta}(\mathbf{q}, \omega). \quad (44)$$

The dynamic structure factor in Eq. (43) contains the elastic scattering, of order  $S^2$  with its  $O(S)$  moment reduction, and the one-magnon scattering, of order  $S$ . The two-magnon continuum from the connected longitudinal correlation is of order  $S^0$  and is not included. At a zero mode, for example a Goldstone mode at a magnetic Bragg vector, the one-magnon weights can diverge as  $\varepsilon_{n\mathbf{q}} \rightarrow 0$ , because the entries of  $T_{\mathbf{q}}$  grow as described in *Paraunitary Diagonalization*.

## Broadening

In LSWT the magnons have infinite lifetime, and the dynamic structure factor consists of delta functions. A finite energy resolution or a phenomenological lifetime is represented by multiplying the correlation function by  $\exp(-\eta|t|)$  with  $\eta > 0$ . Because the particle and hole entries of  $N_{\mathbf{q}}(t)$  oscillate with opposite signs, each delta function is replaced by a Lorentzian centered at its own energy,

$$\delta(\omega - \varepsilon_{n\mathbf{q}}) \rightarrow L_{\eta}(\omega - \varepsilon_{n\mathbf{q}}), \quad \delta(\omega + \varepsilon_{n,-\mathbf{q}}) \rightarrow L_{\eta}(\omega + \varepsilon_{n,-\mathbf{q}}), \quad L_{\eta}(x) = \frac{1}{\pi} \frac{\eta}{x^2 + \eta^2}, \quad (45)$$

and the elastic  $\delta(\omega)$  is replaced by  $L_{\eta}(\omega)$ . The Lorentzian has unit area and full width at half maximum  $2\eta$ , so the frequency integral  $\zeta^{\alpha\beta}(\mathbf{q})$  is unchanged. The broadening is not part of LSWT; other line shapes, such as a Gaussian resolution function, enter in the same way.

## Magnetic Neutron Scattering Intensity

For unpolarized neutrons, the magnetic scattering cross-section contains the dynamic structure factor through the projection onto the components perpendicular to the momentum transfer,

$$\mathcal{S}_\perp(\mathbf{q}, \omega) = \sum_{\alpha, \beta} (\delta_{\alpha\beta} - \hat{q}_\alpha \hat{q}_\beta) \mathcal{S}^{\alpha\beta}(\mathbf{q}, \omega), \quad \hat{\mathbf{q}} = \mathbf{q}/|\mathbf{q}|, \quad (46)$$

up to the magnetic form factor, the  $g$  factor, and constant prefactors. The projector is symmetric in  $(\alpha, \beta)$ , so only the symmetric part of  $\mathcal{S}^{\alpha\beta}$  contributes. For momentum transfer in the plane of a two-dimensional layer,  $\hat{\mathbf{q}} = (q_x, q_y, 0)/|\mathbf{q}|$ . When the  $g$  tensor  $\mathbf{g}_I$  of *Bilinear Spin Hamiltonian* is anisotropic, the neutron couples to the magnetic moment rather than to the spin, and the spin correlations are replaced by moment correlations with the vertex components  $\sum_\gamma g_\mu^{\alpha\gamma} V^\gamma$  for sublattice  $\mu$ .

## Retarded Response and Spectral Function

The linear response of the spin component  $\hat{S}_\mathbf{q}^\alpha$  to a field coupling to  $\hat{S}_{-\mathbf{q}}^\beta$  is given by the retarded function

$$G_R^{\alpha\beta}(\mathbf{q}, \omega) = -i \int_0^\infty dt \exp(i(\omega + i0^+)t) \langle [\hat{S}_\mathbf{q}^\alpha(t), \hat{S}_{-\mathbf{q}}^\beta(0)] \rangle. \quad (47)$$

We define the spectral function as the time Fourier transform of the same commutator,

$$A^{\alpha\beta}(\mathbf{q}, \omega) = \frac{1}{2\pi} \int_{-\infty}^\infty dt \exp(i\omega t) \langle [\hat{S}_\mathbf{q}^\alpha(t), \hat{S}_{-\mathbf{q}}^\beta(0)] \rangle. \quad (48)$$

Three relations connect these functions with each other and with the dynamic structure factor.

**Spectral representation.** Writing the commutator as  $\int d\omega' \exp(-i\omega' t) A^{\alpha\beta}(\mathbf{q}, \omega')$  and performing the time integral gives

$$G_R^{\alpha\beta}(\mathbf{q}, \omega) = \int_{-\infty}^\infty d\omega' \frac{A^{\alpha\beta}(\mathbf{q}, \omega')}{\omega - \omega' + i0^+}.$$

**Fluctuation–dissipation relation.** The first term of the commutator gives  $\mathcal{S}^{\alpha\beta}(\mathbf{q}, \omega)$ . The second term,  $\langle \hat{S}_{-\mathbf{q}}^\beta(0) \hat{S}_\mathbf{q}^\alpha(t) \rangle = C^{\beta\alpha}(-\mathbf{q}, -t)$ , gives  $\mathcal{S}^{\beta\alpha}(-\mathbf{q}, -\omega)$ . With detailed balance,

$$A^{\alpha\beta}(\mathbf{q}, \omega) = \mathcal{S}^{\alpha\beta}(\mathbf{q}, \omega) - \mathcal{S}^{\beta\alpha}(-\mathbf{q}, -\omega) = (1 - \exp(-\beta\omega)) \mathcal{S}^{\alpha\beta}(\mathbf{q}, \omega). \quad (49)$$

**Anti-Hermitian part.** The dynamic structure factor is a Hermitian matrix in  $(\alpha, \beta)$  at each  $\omega$ , and so is  $A^{\alpha\beta}$ . The spectral representation then gives

$$A^{\alpha\beta}(\mathbf{q}, \omega) = -\frac{1}{2\pi i} [G_R^{\alpha\beta}(\mathbf{q}, \omega) - (G_R^{\beta\alpha}(\mathbf{q}, \omega))^*]. \quad (50)$$

For a diagonal component this reduces to  $A^{\alpha\alpha}(\mathbf{q}, \omega) = -\pi^{-1} \text{Im } G_R^{\alpha\alpha}(\mathbf{q}, \omega)$ . For  $\alpha \neq \beta$ , the imaginary part of a single element  $G_R^{\alpha\beta}$  is not the spectral function unless  $G_R$  is symmetric in  $(\alpha, \beta)$ .

## LSWT Result

Inserting Eq. (43) into Eq. (49), the thermal factors cancel:  $(1 - \exp(-\beta\varepsilon))(1 + n_B(\varepsilon)) = 1$  at  $\omega = \varepsilon$ , and  $(1 - \exp(\beta\varepsilon))n_B(\varepsilon) = -1$  at  $\omega = -\varepsilon$ . The elastic part drops out because  $1 - \exp(0) = 0$ . The spectral function and the retarded function are

$$\begin{aligned} A^{\alpha\beta}(\mathbf{q}, \omega) &= \sum_{n=1}^{N_{\text{sub}}} \left[ W_n^{\alpha\beta}(\mathbf{q}) \delta(\omega - \varepsilon_{n\mathbf{q}}) - W_n^{\beta\alpha}(-\mathbf{q}) \delta(\omega + \varepsilon_{n,-\mathbf{q}}) \right], \\ G_R^{\alpha\beta}(\mathbf{q}, \omega) &= \sum_{n=1}^{N_{\text{sub}}} \left[ \frac{W_n^{\alpha\beta}(\mathbf{q})}{\omega - \varepsilon_{n\mathbf{q}} + i0^+} - \frac{W_n^{\beta\alpha}(-\mathbf{q})}{\omega + \varepsilon_{n,-\mathbf{q}} + i0^+} \right]. \end{aligned} \quad (51)$$

In LSWT both functions are independent of temperature, since the magnons do not interact. Replacing  $0^+$  by  $\eta$  in  $G_R$  and taking the anti-Hermitian part replaces the delta functions of  $A^{\alpha\beta}$  by the Lorentzians  $L_\eta$  of Eq. (45).

The frequency integral of the spectral function is the equal-time commutator  $\langle [\hat{S}_{\mathbf{q}}^\alpha, \hat{S}_{-\mathbf{q}}^\beta] \rangle$ . In LSWT, the paraunitarity  $T_{\mathbf{q}} \Sigma_3 T_{\mathbf{q}}^\dagger = \Sigma_3$  turns the sum of the weights into a property of the vertex alone,

$$\int_{-\infty}^{\infty} d\omega A^{\alpha\beta}(\mathbf{q}, \omega) = (V^\alpha T_{\mathbf{q}}) \Sigma_3 (V^\beta T_{\mathbf{q}})^\dagger = V^\alpha \Sigma_3 (V^\beta)^\dagger = \frac{i}{N_{\text{sub}}} \sum_{\mu=1}^{N_{\text{sub}}} S_\mu \sum_{\gamma} \epsilon^{\alpha\beta\gamma} n_\mu^\gamma, \quad (52)$$

where  $\epsilon^{\alpha\beta\gamma}$  is the Levi-Civita symbol. The last step uses  $e_\mu^- \times e_\mu^+ = i n_\mu$ . The result is the spin commutation relation  $[\hat{S}_I^\alpha, \hat{S}_I^\beta] = i \sum_{\gamma} \epsilon^{\alpha\beta\gamma} \hat{S}_I^\gamma$ , averaged over the sites, at leading order in  $S$ . It holds for every  $\mathbf{q}$  and provides a check of the normalization of  $T_{\mathbf{q}}$ .

For Cartesian components, the conjugation relation  $[C^{\alpha\beta}(\mathbf{q}, t)]^* = C^{\beta\alpha}(\mathbf{q}, -t)$  of *Spin Correlations* and stationarity give  $\langle \hat{S}_{-\mathbf{q}}^\beta \hat{S}_{\mathbf{q}}^\alpha(t) \rangle = [C^{\alpha\beta}(-\mathbf{q}, t)]^*$ , so the commutator is  $C^{\alpha\beta}(\mathbf{q}, t) - [C^{\alpha\beta}(-\mathbf{q}, t)]^*$ . It reduces to  $2i \text{Im } C^{\alpha\beta}(\mathbf{q}, t)$  only when  $C^{\alpha\beta}(-\mathbf{q}, t) = C^{\alpha\beta}(\mathbf{q}, t)$ , for example when a symmetry maps  $\mathbf{q}$  to  $-\mathbf{q}$ ; then the retarded function can also be written as  $G_R^{\alpha\beta}(\mathbf{q}, \omega) = 2 \int_0^\infty dt \exp(i(\omega + i0^+)t) \text{Im } C^{\alpha\beta}(\mathbf{q}, t)$ . For a nonreciprocal spectrum,  $\varepsilon_{n\mathbf{q}} \neq \varepsilon_{n,-\mathbf{q}}$ , the commutator must be kept in the general form. For complex combinations of components, the conjugation relation exchanges the components with their complex conjugates, and the commutator must be written out explicitly.

## References

### Internal Documents

- *Spin Correlations*: defines  $C^{\alpha\beta}(\mathbf{q}, t)$ , the spin vertex  $V^\alpha$ , the magnon correlation matrix  $N_{\mathbf{q}}(t)$ , and the elastic correlation.
- *Paraunitary Diagonalization*: supplies  $T_{\mathbf{q}}$ , its particle-hole structure, its gauge freedom, and its behavior at zero modes.
- *Thermodynamics*: defines the Bose-Einstein occupations.
- *Bilinear Spin Hamiltonian*: defines the  $g$  tensor.
- *Magnon Observables*: supplies the reduced moments that enter the elastic part.
- *Notation and Conventions*: distinguishes the external momentum  $\mathbf{q}$  from the magnon momentum  $\mathbf{k}$ .

## External Sources

- G. L. Squires, *Introduction to the Theory of Thermal Neutron Scattering*, 3rd ed. (Cambridge University Press, 2012), [doi:10.1017/CBO9781139107808](https://doi.org/10.1017/CBO9781139107808): magnetic neutron cross-section with the projection  $\delta_{\alpha\beta} - \hat{q}_\alpha \hat{q}_\beta$ , detailed balance, and the fluctuation–dissipation relation.

# Appendix: Luttinger–Tisza Method

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LSWT expands about a classical reference configuration, and the expansion is meaningful only when that configuration minimizes the classical energy. The Luttinger–Tisza (LT) method gives a lower bound on the classical energy of a bilinear spin model by relaxing the fixed length of each spin to a single global constraint. The relaxed problem reduces to the diagonalization of a matrix in momentum space. When a configuration built from the minimizing eigenvectors also has unit spin length at every site, it attains the bound and is a classical ground state. This appendix derives the bound, the conditions for attaining it with a single wave vector, the generalization to inequivalent sublattices, and the limits of the method.

## Classical Energy as a Quadratic Form

We consider the bilinear exchange and single-ion terms of *Bilinear Spin Hamiltonian* without the Zeeman term. Replacing each spin operator by the classical vector  $S_I \mathbf{n}_I$ , with unit vector  $\mathbf{n}_I$ , and writing the link sum as an ordered-pair sum gives the classical energy

$$E_{\text{cl}}[\{\mathbf{n}_I\}] = \frac{1}{2} \sum_{I \neq J} S_I S_J \mathbf{n}_I^T \mathbf{J}_{IJ} \mathbf{n}_J + \sum_I S_I^2 \mathbf{n}_I^T \mathbf{D}_I \mathbf{n}_I,$$

where  $\mathbf{J}_{JI} = \mathbf{J}_{IJ}^T$ . The energy is a quadratic form in the  $3N_{\text{site}}$  components of the unit vectors. The ground-state problem minimizes this form under the strong constraint

$$|\mathbf{n}_I| = 1 \quad \text{for every site } I.$$

The LT method replaces the  $N_{\text{site}}$  strong constraints by the single weak constraint

$$\sum_I |\mathbf{n}_I|^2 = N_{\text{site}},$$

which every configuration satisfying the strong constraint also satisfies.

## Momentum-Space Form

The LT problem uses the unit cell of the Hamiltonian, with sites  $I = (i, a)$  at  $\mathbf{r}_{ia} = \mathbf{R}_i + \boldsymbol{\delta}_a$  and  $a = 1, \dots, N_s$ ; the magnetic unit cell of the ground state is not known in advance. With the full-position Fourier convention of *Momentum-Space BdG Hamiltonian*,

$$n_{ia} = \frac{1}{\sqrt{N_{\text{uc}}}} \sum_{\mathbf{q}} \exp(i\mathbf{q} \cdot \mathbf{r}_{ia}) n_a(\mathbf{q}), \quad n_a(-\mathbf{q}) = n_a(\mathbf{q})^*,$$

where the reality condition follows from real  $n_{ia}$  and the momenta run over the Brillouin zone of the Hamiltonian's unit cell. Translation invariance makes the energy diagonal in  $\mathbf{q}$ :

$$E_{\text{cl}} = \sum_{\mathbf{q}} n(\mathbf{q})^\dagger L_{\mathbf{q}} n(\mathbf{q}), \quad [L_{\mathbf{q}}]_{ab} = \frac{1}{2} \sum_{J \in b} S_a S_b J_{IJ} \exp(i\mathbf{q} \cdot (\mathbf{r}_J - \mathbf{r}_I)) + \delta_{ab} S_a^2 D_a, \quad (53)$$

where  $n(\mathbf{q})$  collects the  $3N_s$  components  $n_a(\mathbf{q})$ ,  $I$  is a fixed site of sublattice  $a$ , and the sum runs over the sites  $J \neq I$  of sublattice  $b$  coupled to  $I$ . Each  $[L_{\mathbf{q}}]_{ab}$  is a  $3 \times 3$  block. The relation  $J_{JI} = J_{IJ}^\dagger$  makes the  $3N_s \times 3N_s$  LT matrix  $L_{\mathbf{q}}$  Hermitian, and real interaction matrices give  $L_{-\mathbf{q}} = L_{\mathbf{q}}^*$ . The spin lengths are absorbed into  $L_{\mathbf{q}}$ , so the constraints refer to unit vectors for any  $S_a$ . The weak constraint becomes  $\sum_{\mathbf{q}} \|n(\mathbf{q})\|^2 = N_{\text{site}}$ .

## Luttinger–Tisza Lower Bound

Let  $\lambda_{\min}(\mathbf{q})$  be the lowest eigenvalue of  $L_{\mathbf{q}}$  and  $\lambda_{\text{LT}} = \min_{\mathbf{q}} \lambda_{\min}(\mathbf{q})$ . Each momentum component satisfies  $n(\mathbf{q})^\dagger L_{\mathbf{q}} n(\mathbf{q}) \geq \lambda_{\text{LT}} \|n(\mathbf{q})\|^2$ , so every configuration obeying the weak constraint, and in particular every configuration of unit spins, satisfies

$$\frac{E_{\text{cl}}}{N_{\text{site}}} \geq \lambda_{\text{LT}}. \quad (54)$$

Equality holds exactly when  $n(\mathbf{q})$  vanishes except at the minimizing wave vectors  $\mathbf{Q}$  and their negatives, and lies in the eigenspace of  $\lambda_{\text{LT}}$  there. A configuration of this form that also satisfies the strong constraint attains the bound and is therefore a classical ground state, with energy  $N_{\text{site}} \lambda_{\text{LT}}$ . If no such configuration exists,  $\lambda_{\text{LT}}$  is a strict lower bound and the method does not determine the ground state.

## Single-Wave-Vector Configurations

The simplest candidates use one minimizing wave vector  $\mathbf{Q}$  with an eigenvector  $\mathbf{w} = (w_1, \dots, w_{N_s})$  of  $\lambda_{\text{LT}}$ . The component at  $-\mathbf{Q}$  is fixed by the reality condition, since  $\mathbf{w}^*$  is an eigenvector of  $L_{-\mathbf{Q}}$ . The configuration is

$$n_{ia} = \text{Re}[u_a \exp(i\mathbf{Q} \cdot \mathbf{R}_i)], \quad u_a = w_a \exp(i\mathbf{Q} \cdot \delta_a),$$

with the cell amplitude  $u_a$  and an overall scale of  $\mathbf{w}$  still free. Its squared length is

$$|n_{ia}|^2 = \frac{1}{2} |u_a|^2 + \frac{1}{2} \text{Re}[(u_a \cdot u_a) \exp(2i\mathbf{Q} \cdot \mathbf{R}_i)], \quad (55)$$

where  $u_a \cdot u_a = \sum_{\gamma} (u_a^{\gamma})^2$  carries no complex conjugation. The phases  $\exp(2i\mathbf{Q} \cdot \mathbf{R}_i)$  over all cells form a subgroup of the unit circle, and the strong constraint takes one of three forms:

- If  $2Q$  is a reciprocal-lattice vector, as at the zone center or at half reciprocal vectors, then  $\exp(iQ \cdot R_i) = \pm 1$ , the configuration is collinear within each sublattice, and the condition is  $|\text{Re } u_a| = 1$  after a suitable choice of the overall phase of  $w$ .
- If  $4Q$  is a reciprocal-lattice vector but  $2Q$  is not, the phases  $\exp(2iQ \cdot R_i)$  take only the values  $\pm 1$ . The condition is  $\text{Re}(u_a \cdot u_a) = 0$  and  $|u_a|^2 = 2$ . Writing  $u_a = x_a + iy_a$ , this requires  $|x_a| = |y_a| = 1$  but leaves the angle between  $x_a$  and  $y_a$  free. The configuration cycles through  $x_a, -y_a, -x_a, y_a$ ; for  $y_a = \pm x_a$  it is the collinear up-up-down-down pattern.
- Otherwise the phases take at least three values that are not confined to  $\pm 1$ , and the condition is  $u_a \cdot u_a = 0$  and  $|u_a|^2 = 2$ . Then  $x_a$  and  $y_a$  are orthonormal, and each sublattice forms a planar spiral in the plane spanned by  $x_a$  and  $y_a$ .

When the eigenspace of  $\lambda_{LT}$  at  $Q$  has more than one dimension,  $w$  may be any vector in it, and the conditions become equations for its coefficients. For a Bravais lattice ( $N_s = 1$ ) with isotropic Heisenberg exchange,  $L_q = S^2 J(q) I_3$  with  $J(q) = \frac{1}{2} \sum_{\Delta} J_{\Delta} \exp(iq \cdot \Delta)$ , where  $\Delta$  runs over all neighbor vectors. Every complex vector is then an eigenvector, a planar spiral always satisfies the strong constraint, and a spiral with wave vector  $Q$  is a ground state, as shown by Lyons and Kaplan. When several inequivalent wave vectors minimize  $J(q)$ , other ground states, including multiple-wave-vector states, can have the same energy. For the triangular-lattice antiferromagnet with nearest-neighbor exchange  $J > 0$ ,  $J(q) = J \sum_{m=1}^3 \cos(q \cdot a_m)$  over the three bond directions has its minimum  $-\frac{3}{2}J$  at the zone corner, and the spiral at that wave vector is the  $120^\circ$  state with energy  $-\frac{3}{2}JS^2$  per site.

## Inequivalent Sublattices and the Generalized Method

For  $N_s > 1$ , the minimizing eigenvector generally distributes its weight unequally among the sublattices,  $|w_a| \neq |w_b|$ , while the strong constraint requires equal lengths. The single weak constraint then allows configurations that no unit-spin state can reach, and the bound is often not attained. A stronger bound follows from one Lagrange multiplier  $\lambda_a$  per sublattice. For unit spins,  $\sum_I \lambda_{a(I)} (|n_I|^2 - 1) = 0$ , so

$$E_{cl} = \sum_q n(q)^\dagger (L_q - \Lambda) n(q) + N_{uc} \sum_{a=1}^{N_s} \lambda_a, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_{N_s}) \otimes I_3.$$

If  $L_q - \Lambda$  is positive semidefinite at every  $q$ , the first term is nonnegative and  $E_{cl} \geq N_{uc} \sum_a \lambda_a$ . The best bound of this form maximizes  $\sum_a \lambda_a$  under that condition. Equal multipliers  $\lambda_a = \lambda_{LT}$  recover Eq. (54), so the generalized bound is never weaker. Lyons and Kaplan introduced this generalization through adjustable parameters in the weak constraint, which allows crystallographically inequivalent spins.

## Scope and Limitations

- **Zeeman field.** The Zeeman term is linear in the spin vectors, so the energy is no longer a homogeneous quadratic form, and the bound above does not include it.
- **Multiple wave vectors.** When no single-wave-vector configuration satisfies the strong constraint, a combination of several minimizing wave vectors may still do so. Failure of the single-wave-vector construction therefore does not show that the bound is unattainable.
- **Degenerate minima.** When the bound is attained, a minimum of  $\lambda_{\min}(q)$  on a line or an extended region, or a degenerate eigenspace beyond that required by a global spin-rotation sym-

metry, signals a degenerate manifold of classical ground states, and the choice among them is not made by the classical energy. The threefold degeneracy of  $L_q$  for isotropic exchange reflects only the global rotation of all spins. When the bound is not attained, the degeneracy of the LT minimum does not determine the degeneracy of the true ground states.

## Relation to Linear Spin-Wave Theory

LSWT requires a reference configuration of unit spins that is stationary under the fixed-length constraint, as described in *Classical Order and Local Frame*. A configuration that attains the LT bound is a global minimum of the zero-field classical energy. Every harmonic fluctuation about it therefore raises or preserves the energy, and its BdG matrix  $H_k$  is at least positive semidefinite, as required in *Paraunitary Diagonalization*. When the bound is not attained,  $\lambda_{LT}$  still bounds the energy of any candidate configuration obtained by numerical minimization from below, and the minimizing wave vectors indicate the magnetic unit cells to consider.

## References

### Internal Documents

- *Bilinear Spin Hamiltonian*: defines the exchange and single-ion matrices and the link counting used in the classical energy.
- *Classical Order and Local Frame*: defines the classical reference configuration.
- *Momentum-Space BdG Hamiltonian*: defines the full-position Fourier convention.
- *Paraunitary Diagonalization*: states the positive-definiteness requirement on the BdG matrix.

### External Sources

- J. M. Luttinger and L. Tisza, “Theory of Dipole Interaction in Crystals,” *Physical Review* **70**, 954–964 (1946), [doi:10.1103/PhysRev.70.954](https://doi.org/10.1103/PhysRev.70.954): representation of spin arrays as vectors in a many-dimensional space and reduction of the minimum-energy problem to a matrix diagonalization.
- D. H. Lyons and T. A. Kaplan, “Method for Determining Ground-State Spin Configurations,” *Physical Review* **120**, 1580 (1960), [doi:10.1103/PhysRev.120.1580](https://doi.org/10.1103/PhysRev.120.1580): generalization with adjustable parameters in the weak constraint for inequivalent spins; spiral ground state for lattices of equivalent spins.